Networking Fundamentals for Cybersecurity Professionals | by Naem Azam Chowdhury | Medium

The Data Link Layer and Local Area Networks

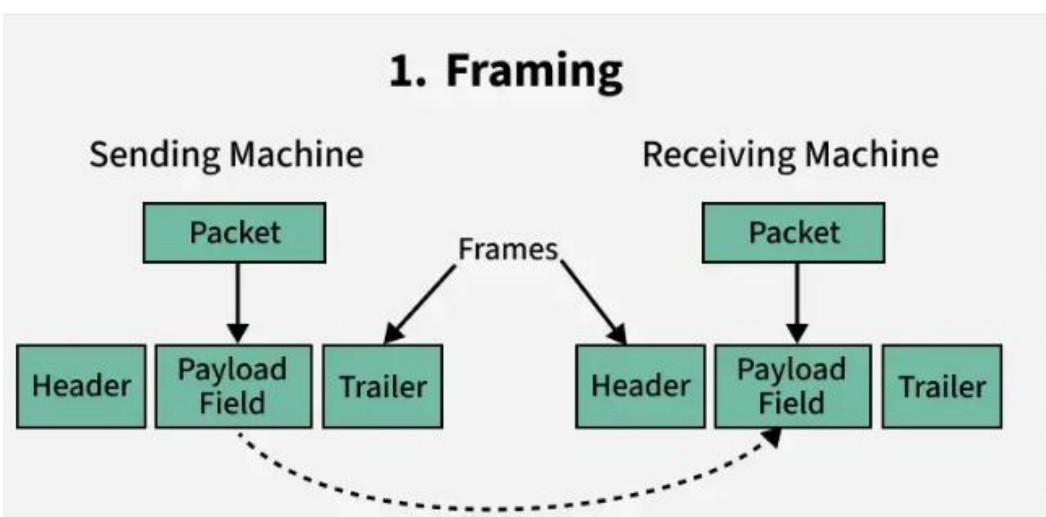
- The data link layer is the second layer from the bottom in the OSI (Open System Interconnection) network architecture model.
- Responsible for the node-to-node delivery of data within the same local network.
- Major role is to ensure error-free transmission of information.
- Also responsible for encoding, decoding, and organizing the outgoing and incoming data.
- Considered as the most complex layer of the OSI model as it hides all the underlying complexities of the hardware from the other above layers.

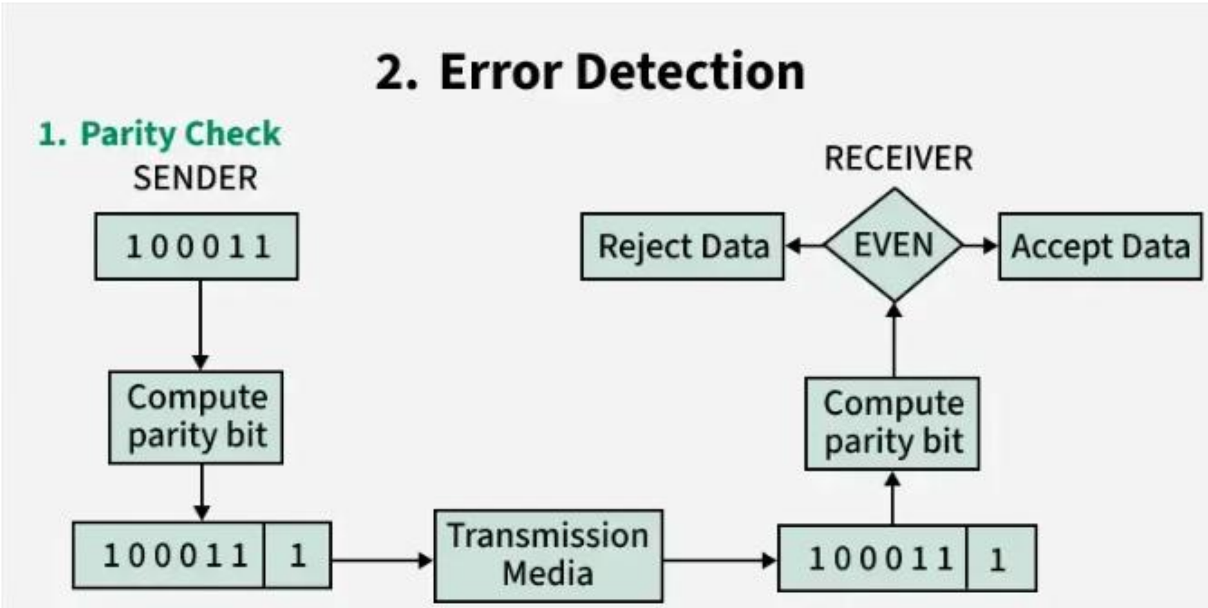
Sub-Layers of The Data Link Layer

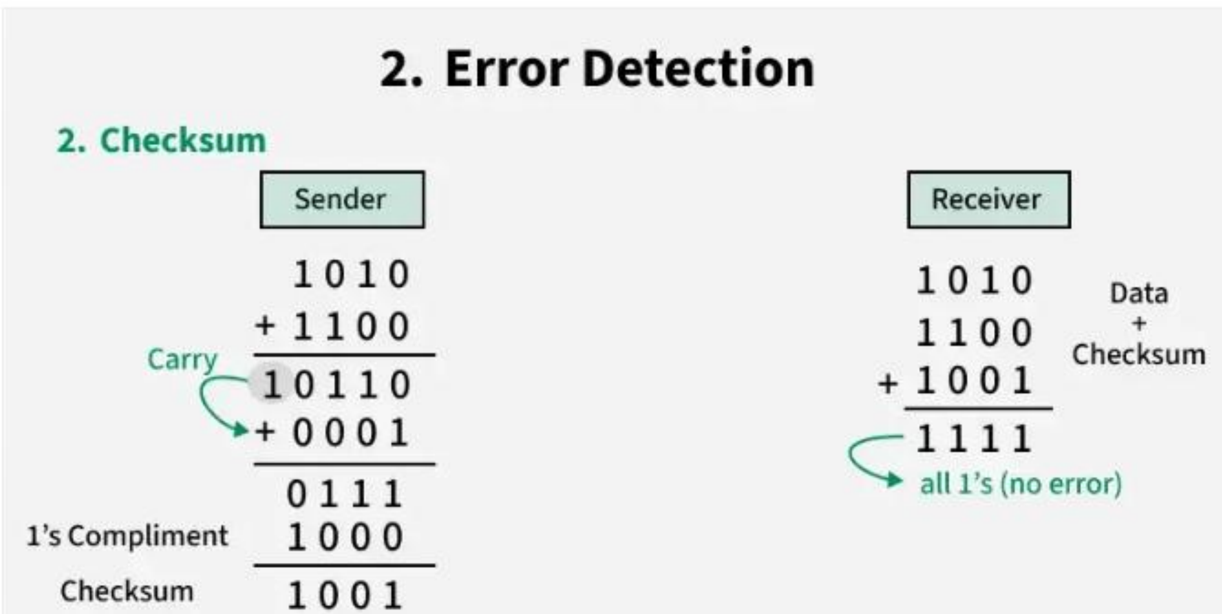
The data link layer is further divided into two sub-layers, which are as follows:

1. **Logical Link Control (LLC):** This sublayer of the data link layer deals with multiplexing, the flow of data among applications and other services, and LLC is responsible for providing error messages and acknowledgments as well.
2. **Media Access Control (MAC):** MAC sublayer manages the device's interaction, responsible for addressing frames, and also controls physical media access. The data link layer receives the information in the form of packets from the Network layer, it divides packets into frames and sends those frames bit-by-bit to the underlying physical layer.

Functions of The Data-link Layer

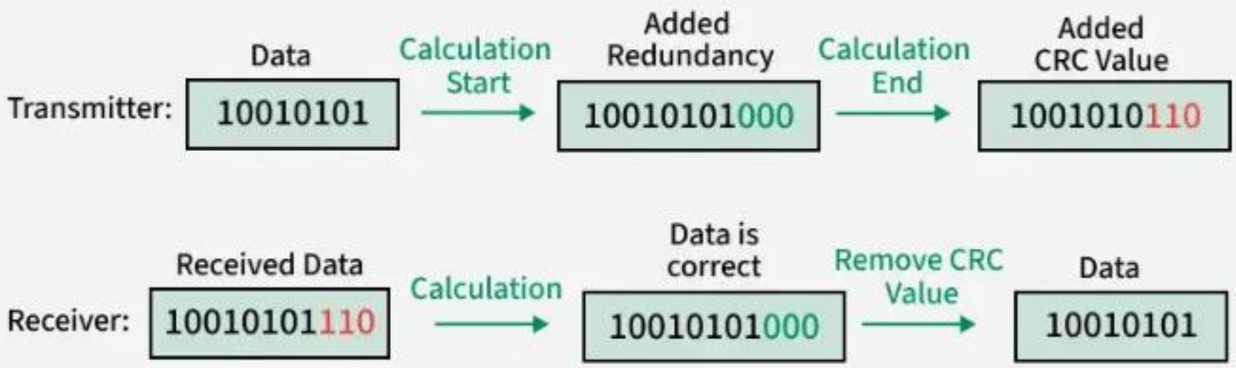



2. Error Detection
1. Parity Check
SENDER
100011
Compute parity bit
100011 1
Transmission Media
RECEIVER
100011 1
Compute parity bit
EVEN
Reject Data
Accept Data


2. Error Detection
2. Checksum
Sender
1010
+ 1100
Carry
10110
+ 0001
0111
1's Compliment
1000
Checksum
1001
Receiver
1010
1100
+ 1001
Data + Checksum
1111
all 1's (no error)

2. Error Detection

3. Cyclic Redundancy Check



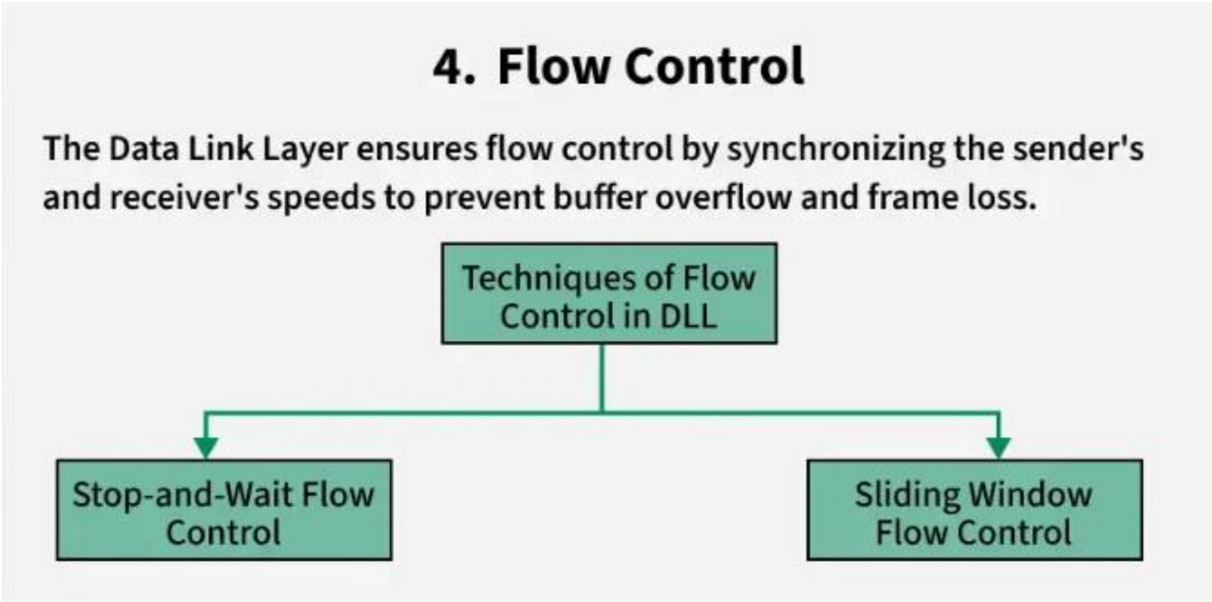
3. Error Correction

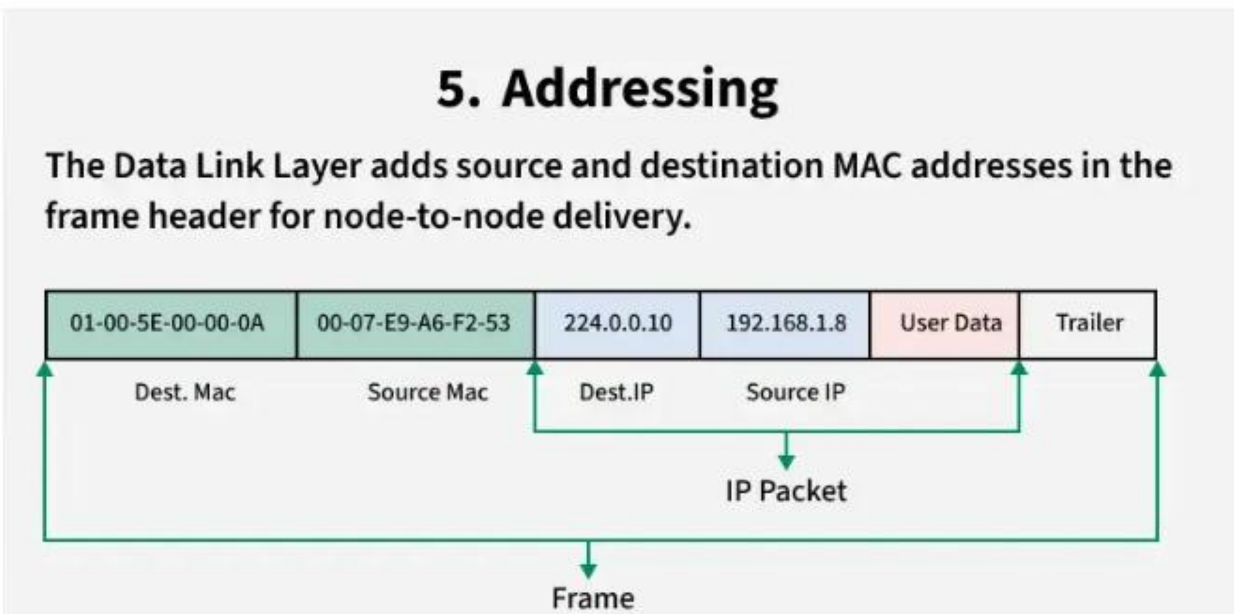
- **Automatic-repeat request (ARQ)**

  ARQ is a feedback-based method where the receiver requests the sender to retransmit the data if errors are detected.

- **Forward error correction (FEC):**

  FEC is a method where the sender includes additional redundant data, allowing the receiver to detect and correct errors without retransmission.





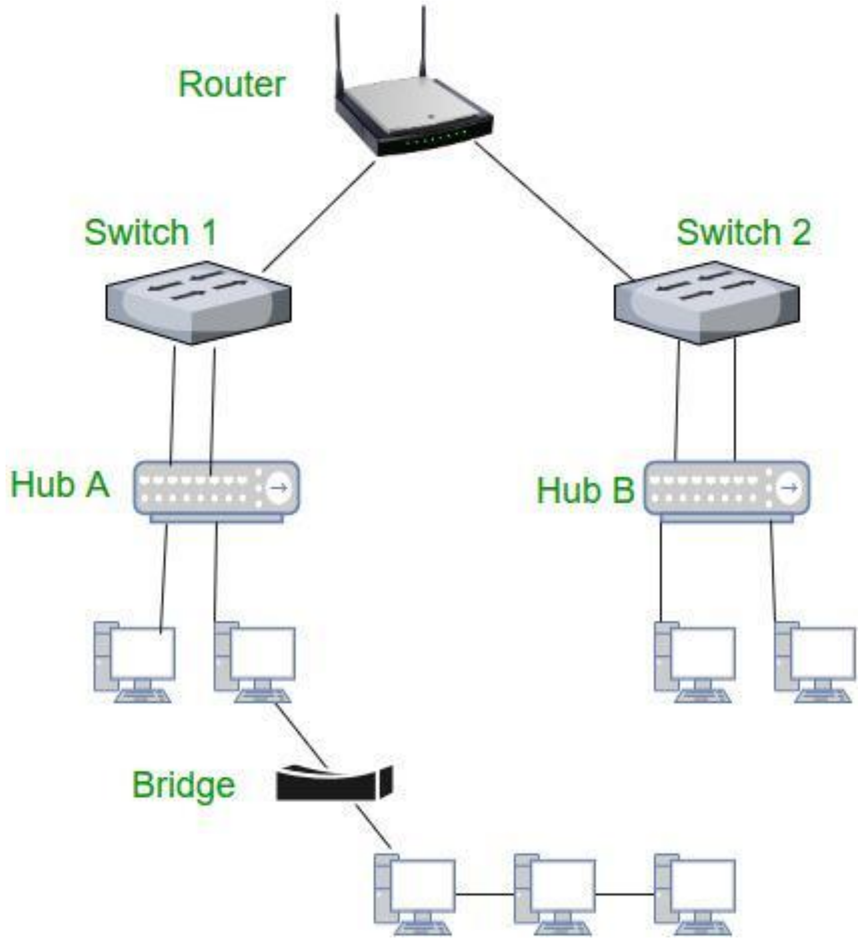
Protocols in Data link layer

There are various protocols in the data link layer, which are as follows:

- Synchronous Data Link Protocol (SDLC)
- High-Level Data Link Protocol (HDLC)
- Serial Line Interface Protocol (SLIP)
- Point to Point Protocol (PPP)
- Link Access Procedure (LAP)
- Link Control Protocol (LCP)
- Network Control Protocol (NCP)

Devices Operating at the Data Link Layer

All these devices rely on MAC addresses for efficient frame delivery and play a crucial role in local network communication and access control.



Devices at Data Link Layer

1. Switch

- A switch is a key device in the Data Link Layer.
- It uses MAC addresses to forward data frames to the correct device within a network.
- Works in local area networks (LANs) to connect multiple devices.

2. Bridge

- A bridge connects two or more LANs, creating a single, unified network.
- Operates at the Data Link Layer by forwarding frames based on MAC addresses.
- Used to reduce network traffic and segment a network.

3. Network Interface Card (NIC)

- A NIC is a hardware component in devices like computers and printers.
- Responsible for adding the MAC address to frames and ensuring proper communication with the network.
- Operates at the Data Link Layer by preparing and sending frames over the physical medium.

4. Wireless Access Point (WAP)

- A WAP allows wireless devices to connect to a wired network.
- Operates at the Data Link Layer by managing wireless MAC addresses.
- Uses protocols like Wi-Fi (IEEE 802.11) to communicate with devices.

5. Layer 2 Switches

- These are specialized switches that only operate at Layer 2, unlike multi-layer switches.
- Responsible for frame forwarding using MAC address tables.

Applications of Data Link Layer

- **Local Area Networks (LANs)**: Enables reliable communication between devices within a local network using protocols like Ethernet (IEEE 802.3).
- **Wireless Networks (Wi-Fi)**: Manages communication between devices in wireless networks via protocols like IEEE 802.11 hence, handling media access and error control.
- **Switches and MAC Addressing**: Facilitates the operation of switches by using MAC addresses to forward data frames to the correct device within the network.
- **Point-to-Point Connections**: Used in protocols like PPP (Point-to-Point Protocol) to establish and manage direct communication between two nodes.

2.1.1. DATA LINK LAYER DESIGN ISSUES

The following are the data link layer design issues

1. Services Provided to the Network Layer

The network layer wants to be able to send packets to its neighbors without worrying about the details of getting it there in one piece.

2. Framing

Group the physical layer bit stream into units called frames. Frames are nothing more than "packets" or "messages". By convention, we use the term "frames" when discussing DLL.

3. Error Control

Sender checksums the frame and transmits checksum together with data. Receiver re-computes the checksum and compares it with the received value.

4. Flow Control

Prevent a fast sender from overwhelming a slower receiver.

Services Provided to the Network Layer

The function of the data link layer is to provide services to the network layer. The principal service is transferring data from the network layer on the source machine to the network layer on the destination machine.

The data link layer can be designed to offer various services. The actual services offered can vary from system to system. Three reasonable possibilities that are commonly provided are

1) **Unacknowledged Connectionless service**

2) **Acknowledged Connectionless service**

3) **Acknowledged Connection-Oriented service**

Error detection and correction in computer networks

Error detection and correction in computer networks is a set of methods that help devices deliver the correct data, even when the path is not perfectly reliable. The sender adds a small amount of extra information. That extra information lets the receiver do one of two things:

• **Detect** that something went wrong (and ask for a resend).

• **Correct** the error (without asking for a resend).

These ideas sound simple, but the design choices are not. Each method trades speed, bandwidth, compute cost, and reliability in a different way.

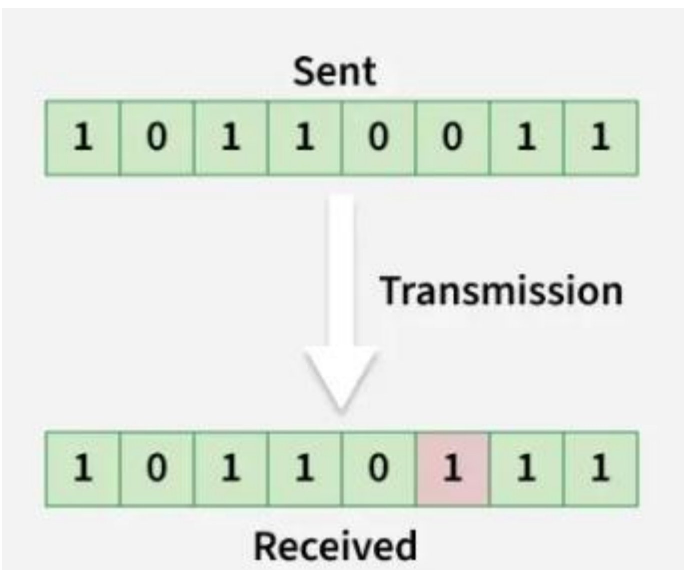
Error Detection in Computer Networks

Error detection is about finding corruption. It does not necessarily fix it on its own. The receiver checks the incoming data using a rule. If the check fails, the receiver knows the data is unreliable.

Types of Errors

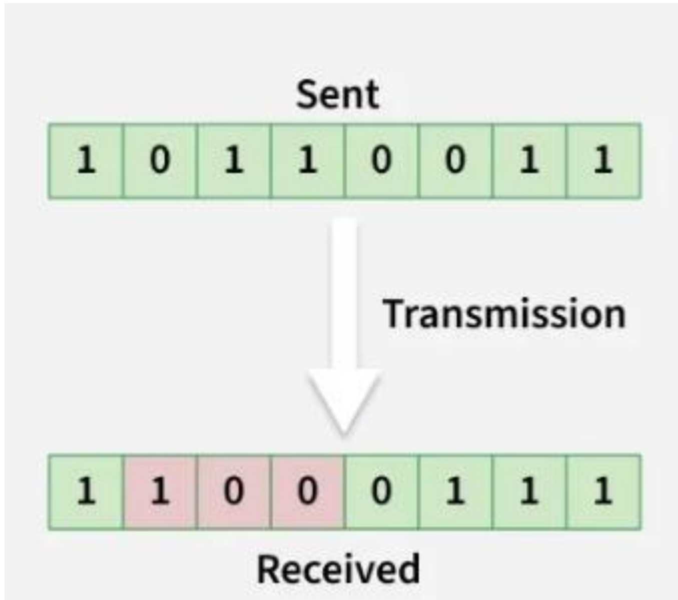
1. Single-Bit Error

A single-bit error occurs when **only one bit** of the transmitted data unit is altered during transmission, resulting in corrupted data.



2. Burst Error

A burst error occurs when two or more consecutive bits in a data unit are corrupted during transmission.



Error Detection Methods

To detect errors, a common technique is to introduce redundancy bits that provide additional information. Various techniques for error detection include:

- Simple Parity Check
- Two-Dimensional Parity Check
- Checksum
- Cyclic Redundancy Check (CRC)

1. Simple Parity Check

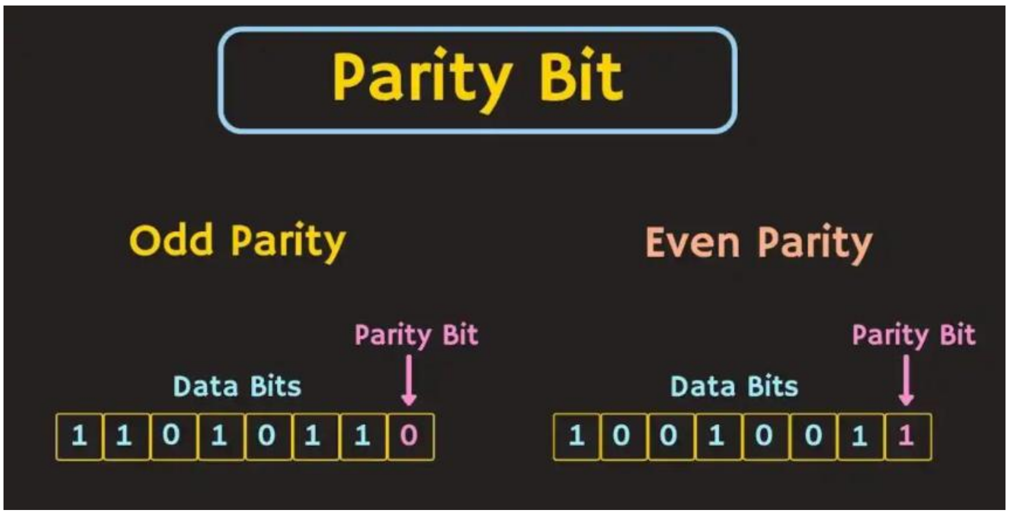
Simple bit parity is a basic error detection technique in which an extra bit, called a parity bit, is added to a data unit before transmission. This parity bit helps the receiver determine whether the transmitted data has been corrupted.

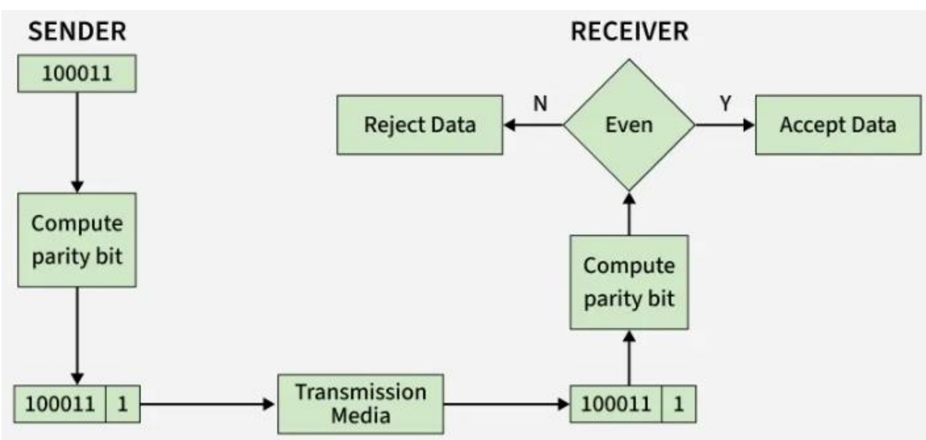
- Detects all single-bit errors in transmitted data.
- Detects any odd number of bit errors, since odd bit changes alter the parity.

- Easy to implement in both hardware and software.
- Fails to detect errors when an even number of bits are corrupted.
- Not suitable for noisy communication channels with frequent errors.

Parity bits are used to maintain a specific parity condition in the data. There are two types of parity:

- **Even Parity**: The parity bit is set so that the total number of 1s in the data (including the parity bit) is even.
- **Odd Parity**: The parity bit is set so that the total number of 1s in the data (including the parity bit) is odd.



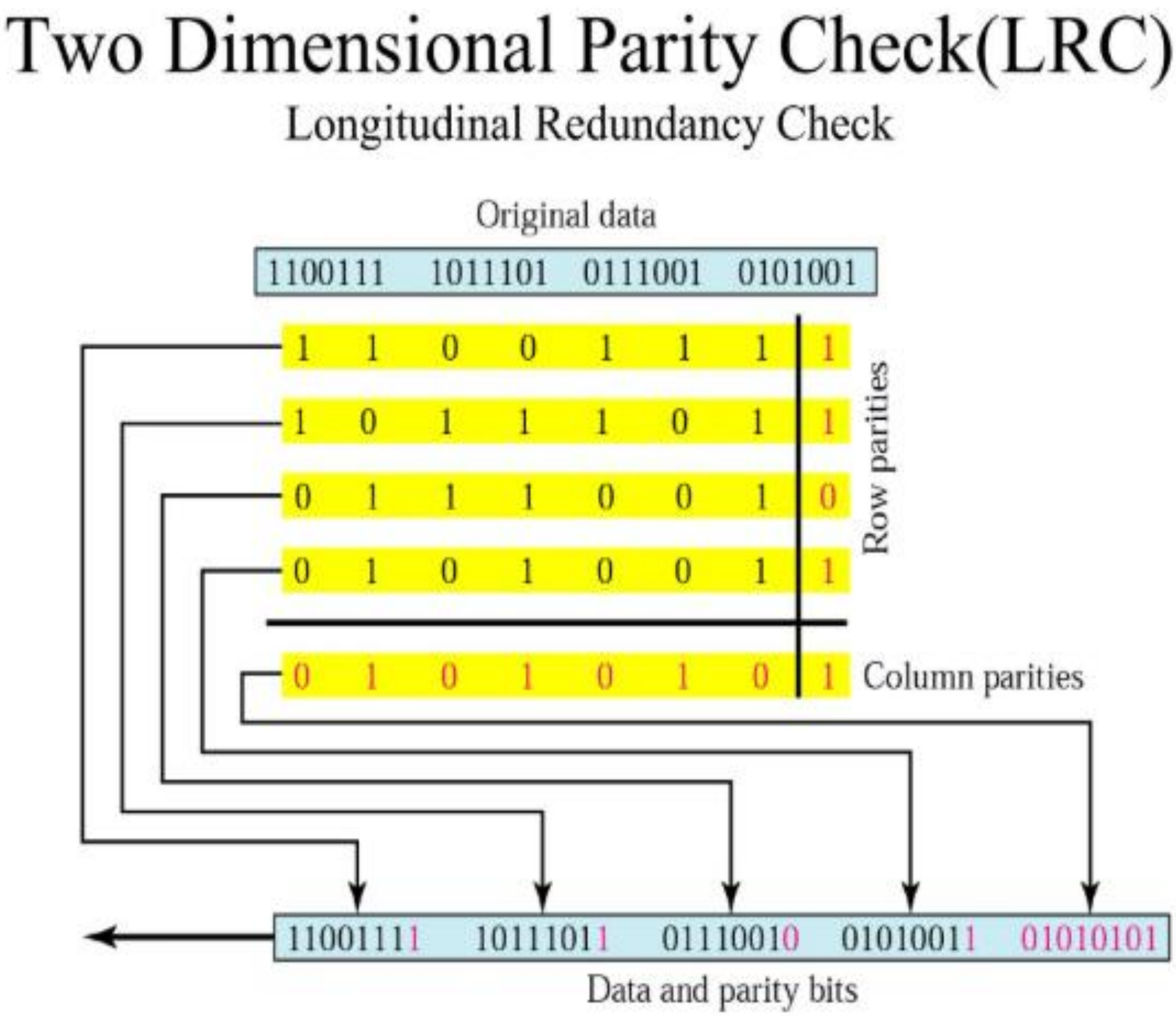


This scheme makes the total number of 1’s even, that is why it is called even parity checking.

2. Two-Dimensional Parity Check

In two-dimensional parity check, parity bits are calculated for each row, similar to a simple parity check. In addition, parity bits are also computed for each column. These row and column parity bits are transmitted along with the data. At the receiver, parity bits are recalculated and compared with the received parity bits to detect errors.

- Can detect and correct all single-bit errors by identifying the exact row and column where the error occurred.
- Can detect many multiple-bit errors occurring at different positions in the data matrix.
- Provides better error detection capability than simple parity check.
- Certain patterns of multiple-bit errors may remain undetected.
- If parity bits themselves are corrupted, error detection may fail.



3. Checksum

Checksum is an error detection technique used to detect errors in transmitted data. In this method, the sender divides the data into fixed-size segments and computes a checksum using 1’s complement arithmetic. The computed checksum is transmitted along with the data. At the receiver, the same computation is performed to verify data integrity.

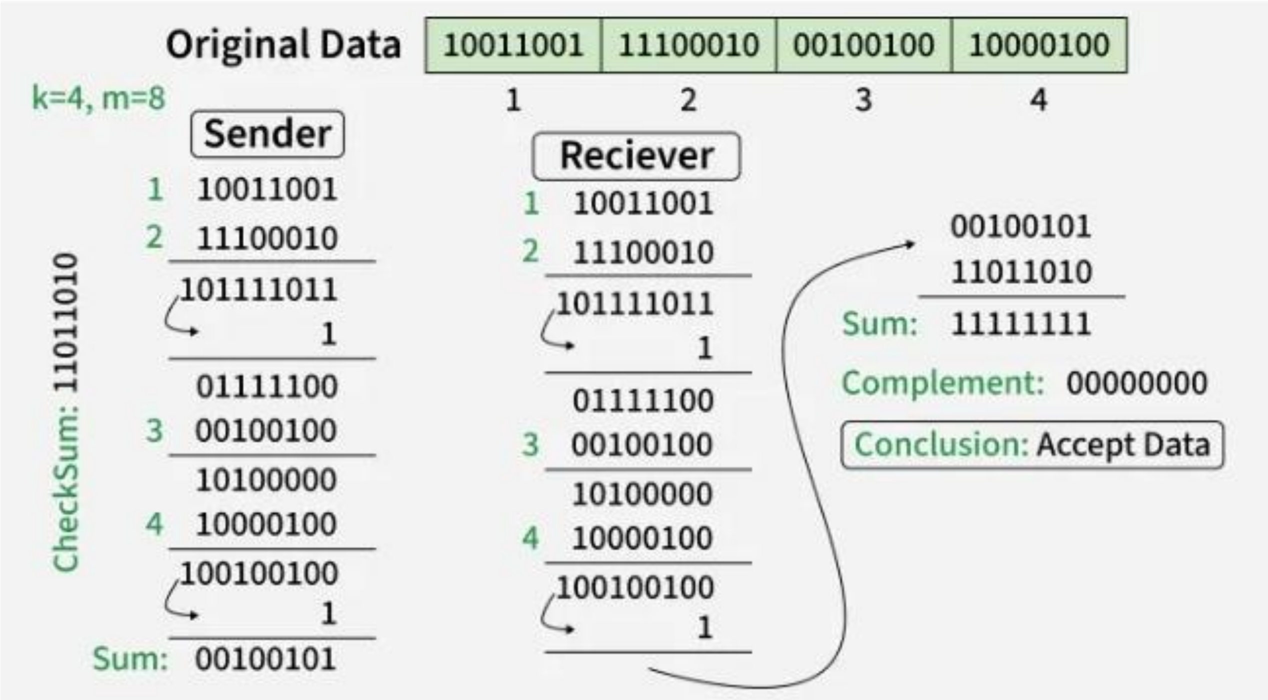
- Can detect many types of errors, including single-bit and multiple-bit errors.
- Provides better error detection than simple parity and two-dimensional parity checks.
- Widely used in networking protocols such as IP, TCP, and UDP.
- Fast to compute and easy to implement in software.
- Supports only error detection and does not provide error correction.
- Some error patterns may go undetected, especially when errors cancel each other during addition.
- Less reliable compared to advanced techniques like Cyclic Redundancy Check (CRC).

Checksum - Operation at Sender's Side

- The data is divided into k segments, each of m bits.
- All segments are added using 1’s complement arithmetic.
- The 1’s complement of the final sum is calculated to generate the checksum.
- The checksum is appended to the data and transmitted to the receiver.

Checksum - Operation at Receiver's Side

- All received segments, including the checksum, are added using 1’s complement arithmetic.
- If the final result is all 1s, the data is considered error-free.
- Otherwise, the data is discarded, indicating that an error has occurred.



****** Checksum (in Computer Networks)** is an **error detection technique** used to verify the integrity of data during transmission.

1. The sender divides data into fixed-size segments (e.g., 16 bits).
2. All segments are added using **1's complement arithmetic**.
3. The final sum is complemented to produce the **checksum**, which is sent along with the data.
4. The receiver repeats the same process on received data + checksum:
   1. If the result is **all 1s (no error)** → data is accepted
   2. Otherwise → **error detected**

Suppose sender wants to send two data blocks:

- Data 1 = 1010
- Data 2 = 1100

Step 1: Add the data (binary addition)
  1010
+ 1100

 10110

Since we are using **4 bits**, take only last 4 bits and add carry:

- Result = 0110
- Carry = 1

Add carry:

0110 + 0001 = 0111

Step 2: Take 1's complement (invert bits)
Checksum = 1000

Step 3: Send data + checksum

Sender sends:

1010, 1100, 1000

Step 4: Receiver side (verification)

Add all:

1010 + 1100 = 0111 (after carry)
0111 + 1000 = 1111

Result

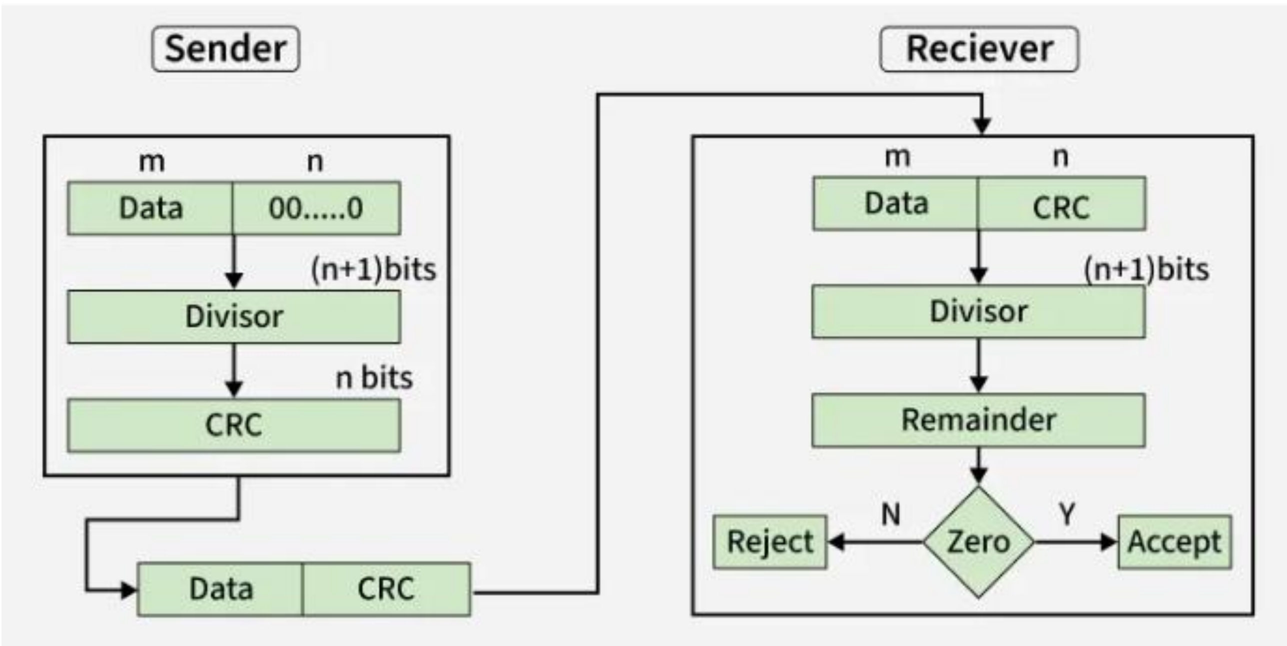
Result = 1111 (all 1s) → **No error**

If not all 1s → **Error detected**

4. Cyclic Redundancy Check (CRC)

- CRC is based on binary division, unlike checksum which uses addition; redundant CRC bits are appended so the transmitted data becomes exactly divisible by a predefined generator polynomial.
- At the receiver, the data is divided using the same polynomial; a zero remainder means the data is accepted, while a non-zero remainder indicates transmission errors.
- Highly effective in detecting single-bit, multiple-bit, and burst errors, making it more reliable than parity checks and checksum methods.

- Widely used in communication protocols such as Ethernet, HDLC, and USB due to its strong error detection capability.
- Provides only error detection, not error correction; its effectiveness depends on the chosen generator polynomial.
- More computationally complex than simple parity or checksum methods and introduces additional processing overhead.



CRC Working

We have given dataword of length m +n and divisor of length k.

Step 1: Append (k-1) zero's to the original message
Step 2: Perform modulo 2 division
Step 3: Remainder of division = CRC
Step 4: Code word = Data with append k-1 zero's + CRC
Note:

- CRC must be k-1 bits
- Length of Code word = n+k-1 bits

OR

1. Take data
2. Divide by a **generator polynomial**
3. Remainder = **CRC bits**
4. Send data + CRC

At Receiver:

- Divide again
- If remainder = 0 → No error
- Else → Error

Advantage:

- Very **powerful and accurate**
- Detects **burst errors (multiple bits)**

Comparison:

Method	Easy Idea	Error Detection Power
Parity	Count 1's	Low
2D Parity	Rows + Columns	Medium
Checksum	Add numbers	Medium
CRC	Divide & remainder	High

Error Correction in Computer Networks

Error correction goes one step further. It aims to fix errors at the receiver side. To do that, the sender adds more redundancy than detection alone. That redundancy lets the receiver reconstruct the original bits even when some bits flip.

There are two broad ways networks "correct" issues:

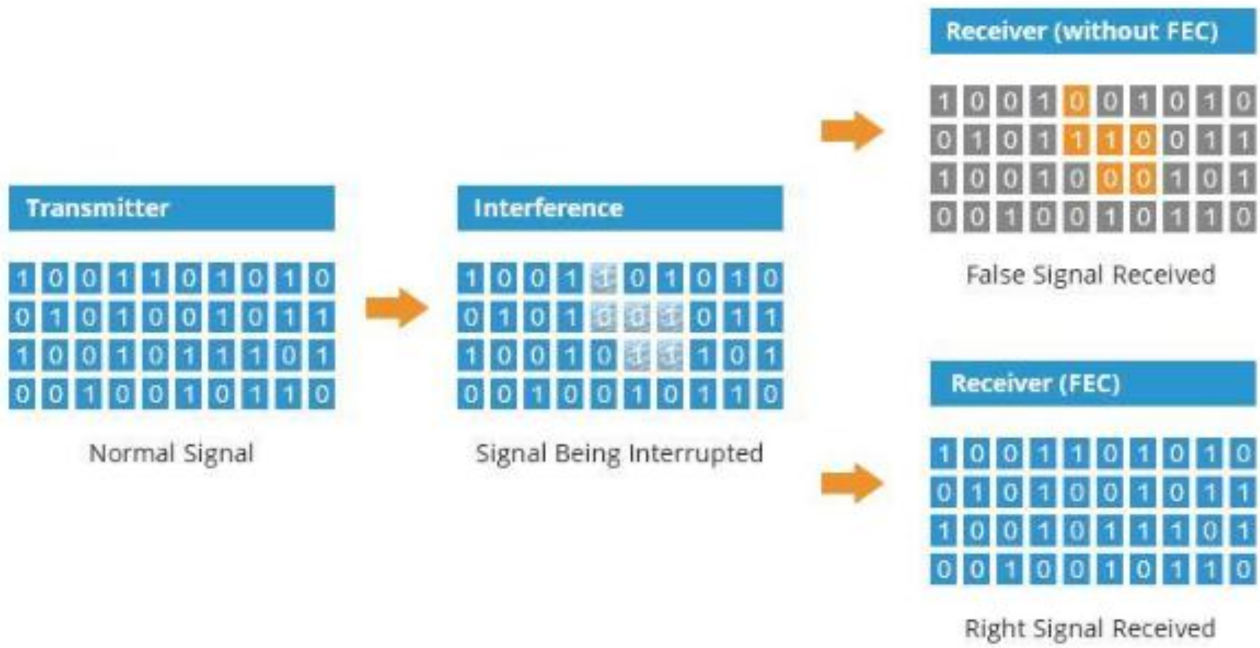
1. Retransmission-based correction (ARQ)

The receiver detects an error, then asks for the data again.

This is not "bit-level correction," but it corrects the delivered result.

2. Forward Error Correction (FEC)

The sender includes enough extra bits so the receiver can correct errors without asking again.



Sender adds **extra redundant bits** so receiver can detect & correct errors **on its own**.

- Sender → adds extra bits
- Receiver → uses them to fix errors
- Example:

If data = 1011
Sender sends = 1011 + extra bits

Advantages:

- No need to resend data
- Faster (useful in real-time systems)

Disadvantages:

- Extra bandwidth required

Used in:

- Satellite communication
- Mobile networks
- Wi-Fi

Common FEC families include:

1. **Hamming codes**: Great for small blocks and single-bit correction. Very educational and still used in some places.

Add parity bits at specific positions to **detect and correct single-bit errors**

Key Concept:

- Positions: 1, 2, 4, 8… are parity bits
- Helps locate exact error position

Advantage:

- Can **correct 1-bit error**

2. **Reed–Solomon codes**: Strong against burst errors (many wrong bits in a row). Used in storage, QR codes, and also in some communication systems.

Works on blocks of data to correct multiple errors

Advantage:

- **Handles** burst errors (many bits)

Used in:

- **CDs, DVDs**
- **QR codes**
- **Digital TV**

3. **Convolutional codes, LDPC, Turbo codes**: Heavy hitters in wireless and high-speed systems. These are advanced coding

techniques used in communication systems to enhance error correction and data transmission reliability but need more computing power.

Data is encoded using previous bits (memory-based)

Feature:

- Continuous encoding
- Uses algorithms like **Viterbi decoding**

Used in:

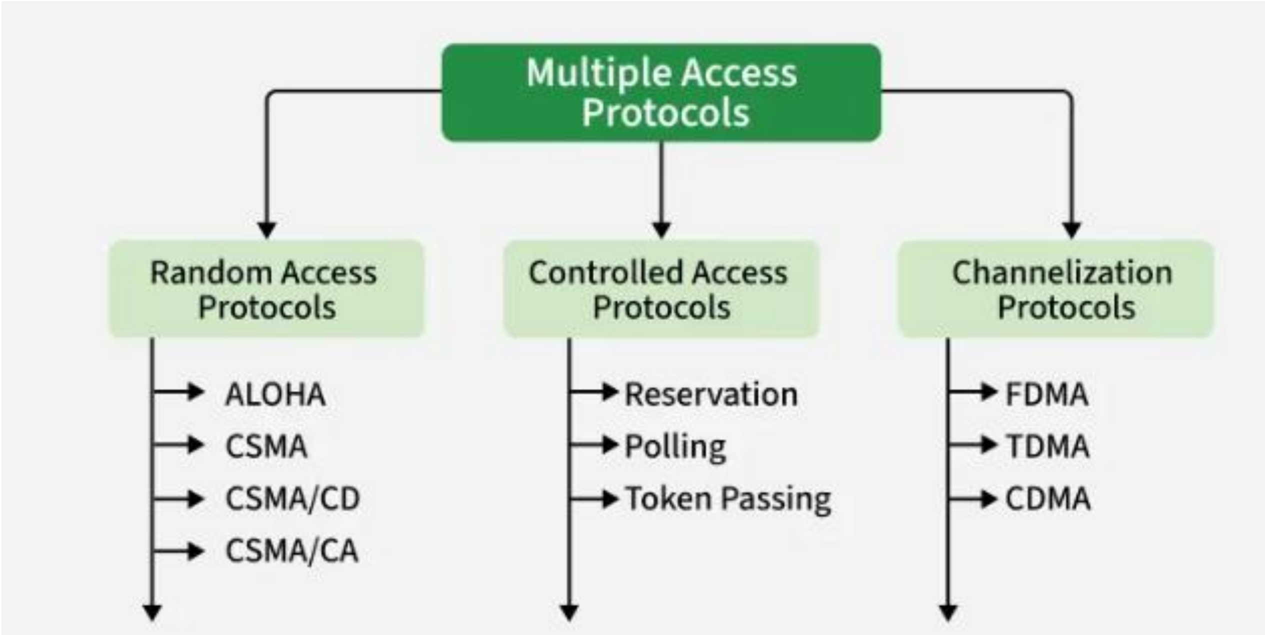
- Wireless communication
- Satellite systems

So when people talk about error detection and correction in computer networks, correction usually means a mix of ARQ + FEC, chosen based on latency and channel quality.

Feature	Error Detection	Error Correction
Purpose	Find error	Find + Fix error
Example	Parity, CRC	Hamming, Reed-Solomon
Retransmission	Needed	Not always needed

Multiple Access Protocols and LANs

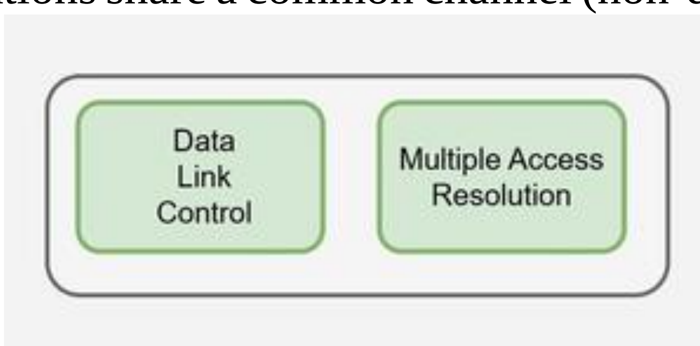
Multiple Access Protocols are a set of rules used in computer networks to control how multiple devices share and access a common communication channel. Since simultaneous transmissions by multiple devices can lead to data collisions and loss, these protocols ensure orderly, fair, and efficient use of the shared medium by coordinating when and how each device can transmit data.



Role in the OSI Model

The Data Link Layer is responsible for the transmission of data between nodes. It has two major functions:

1. **Data Link Control (DLC):** Ensures reliable transmission using framing, error control flow control (e.g., Stop-and-Wait ARQ).
2. **Multiple Access Control (MAC):** Manages access when multiple stations share a common channel (non-dedicated link).

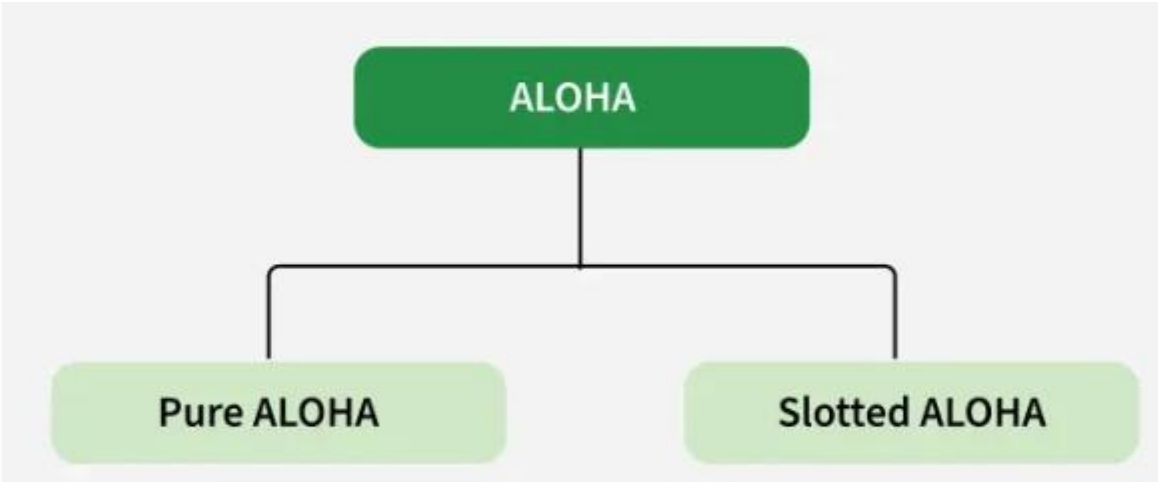


Data Link Layer Functions

1. ALOHA

- Designed for wireless LANs and shared media.

- Multiple stations can transmit simultaneously, leading to collisions.



A

Working:

- A node sends data whenever it has data
- If collision occurs → data is retransmitted after random delay

Types:

- **Pure ALOHA** → No time synchronization
- **Slotted ALOHA** → Time divided into slots

Efficiency:

- Pure ALOHA → 18%
- Slotted ALOHA → 36%

Limitation:

- High collision probability

Pure ALOHA

The mode of random access in which users can transmit at any time is called pure Aloha. This technique is explained below in a stepwise manner.

Step 1 In pure ALOHA, the nodes transmit frames whenever there is data to send.

Step 2 When two or more nodes transmit data simultaneously, then there is a chance of collision and the frames are destroyed.

Step 3 In pure ALOHA, the sender will expect acknowledgement from the receiver.

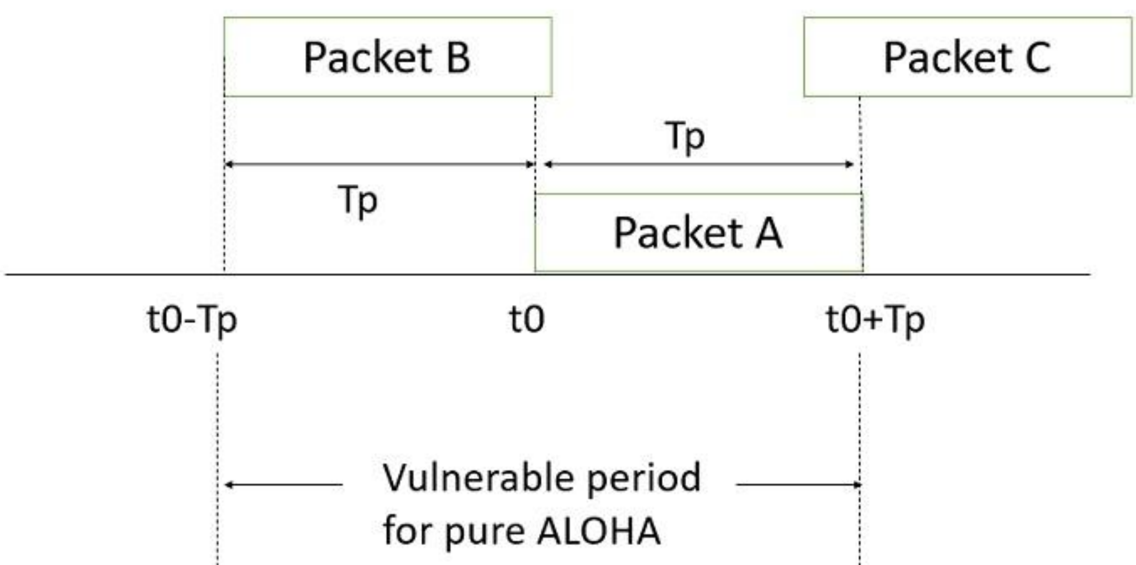
Step 4 If acknowledgement is not received within specified time, the sender node assumes that the frame has been destroyed.

Step 5 If the frame is destroyed by collision the node waits for a random amount of time and sends it again. This waiting time may be random otherwise the same frames will collide multiple times.

Step 6 Therefore, pure ALOHA says that when the time-out period passes, each station must wait for a random amount of time before re-sending its frame. This randomness will help avoid more collisions.

In PURE ALOHA, the vulnerable period is two slot times.

Vulnerable period is the maximum interval over which two packets can overlap and destroy each other. This phenomenon is shown the below figure



The throughput of pure ALOHA is given by the following

Probability of success (PSuccess) is equal to probability of no other packet transmission occurring within the vulnerable period.

Therefore, **Throughput(s)** is defined as the successfully transmitted traffic load and G is the total offered channel traffic load.

Assume that traffic generated for transmission obeys a Poisson distribution,

P (no other packet transmission occurs) = exp (-TG)

Where T is vulnerable period,

Psuccess = exp(-TG)

$$P_{success} = \exp(-TG)$$
$$= S/G$$
$$S = Ge^{-TG}$$

If vulnerable time is 2

$$S_{max} = 1/2e$$
$$= 0.184$$

= 0.184

It means, in Pure ALOHA the channel utilization is 18 percent.

2. Slotted Aloha

Slotted Aloha is a modified version of pure aloha where we divide the time into slots of Tfr. Thus, If a station fails to send the frame in one slot it can't send throughout that slot and have to wait for the next slot beginning. This successfully reduces the vulnerable time down to Tfr which was previously **2* Tfr** for pure aloha. Throughput also increases and turns to be **G*e-G**, where max throughput becomes **.368 when G=1**

2. CSMA (Carrier Sense Multiple Access)

Carrier Sense Multiple Access (CSMA) is a network protocol for carrier transmission that operates in the Medium Access Control (MAC) layer. It senses or listens whether the shared channel for transmission is busy or not, and transmits if the channel is not busy. Using CMSA protocols, more than one users or nodes send and receive data through a shared medium that may be a single cable or optical fiber connecting multiple nodes, or a portion of the wireless spectrum.

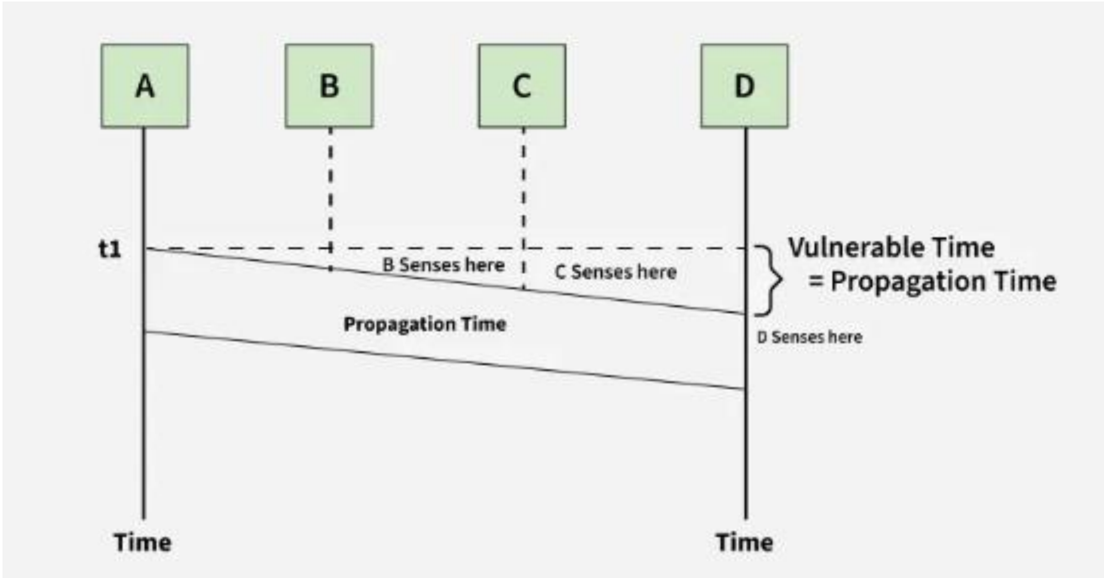
Working Principle

When a station has frames to transmit, it attempts to detect presence of the carrier signal from the other nodes connected to the shared channel. If a carrier signal is detected, it implies that a transmission is in progress. The station waits till the ongoing transmission executes to completion, and then initiates its own transmission. Generally, transmissions by the node are received by all other nodes connected to the channel.

Since, the nodes detect for a transmission before sending their own frames, collision of frames is reduced. However, if two nodes detect an idle channel at the same time, they may simultaneously initiate transmission. This would cause the frames to garble resulting in a collision.

Before transmitting, stations sense the medium:

- If idle -> transmit.
- If busy -> wait.



Types of CSMA:

- CSMA/CD (Collision Detection)
- CSMA/CA (Collision Avoidance)

CSMA/CD (Collision Detection)

CSMA/CD is an extension of CSMA used primarily in wired networks like Ethernet. It includes mechanisms for detecting collisions and taking corrective actions. The key steps in CSMA/CD are:

- **Carrier Sensing:** Devices monitor the channel to check if it is free before transmitting.
- **Transmission:** If the channel is free, the device starts transmitting data.
- **Collision Detection:** Devices continue to monitor the channel while transmitting. If a collision is detected (by sensing interference or a drop in signal quality), the device immediately stops transmitting.
- **Jamming Signal:** The device sends a jamming signal to inform other devices of the collision.
- **Backoff Algorithm:** After a collision, devices wait for a random period before attempting to retransmit. The waiting

time typically increases exponentially with each subsequent collision.

CSMA/CA (Collision Avoidance)

- CSMA/CA is predominantly used in wireless networks, such as Wi-Fi, where collision detection is more challenging due to the nature of wireless communication. Instead of detecting collisions, CSMA/CA focuses on avoiding them. The key steps are:
- **Carrier Sensing:** Devices listen to the channel before transmitting.
- **Collision Avoidance:** If the channel is busy, the device waits for a random backoff period before attempting to sense the channel again. Some implementations use a Request to Send (RTS) and Clear to Send (CTS) mechanism to reserve the channel for a specific communication.
- **Transmission:** If the channel is free, the device transmits the data.
- Advantages of CSMA
- CSMA offers several benefits, making it a popular choice for managing shared communication channels:
- **Efficiency:** By allowing devices to listen to the channel before transmitting, CSMA reduces the likelihood of collisions and improves the overall efficiency of the network.
- **Simplicity:** The protocol is relatively simple to implement, making it suitable for a wide range of network types and sizes.
- **Scalability:** CSMA can accommodate a varying number of devices, from small local networks to large-scale deployments.

Controlled Access Protocols

Here, devices take turns accessing the channel, preventing collisions. For example:

- **Reservation:** Stations reserve slots before transmission.
- **Polling:** Central controller polls each device in turn.
- **Token Passing:** A token (special packet) is passed; only the device holding the token can transmit.

Channelization Protocols

Bandwidth is divided and allocated so that multiple users can transmit simultaneously.

1. Frequency Division Multiple Access (FDMA)

- Bandwidth is divided into frequency bands.
- Guard bands prevent overlap.

2. Time Division Multiple Access (TDMA)

- Time is divided into slots.
- Each station transmits only in its assigned slot.
- Needs synchronization.

3. Code Division Multiple Access (CDMA)

- All stations transmit simultaneously using unique codes.
- Widely used in mobile communication.

4. Orthogonal Frequency Division Multiple Access (OFDMA)

- Bandwidth divided into small subcarriers.
- Widely used in 4G/5G for high data rates.
- High throughput supports multimedia.
- Complex design, high power requirements.

5. Spatial Division Multiple Access (SDMA)

- Uses multiple antennas (MIMO).

- Separates users spatially.
- Improves data rate and signal quality.

Features of Multiple Access Protocols

- **Contention-Based Access:** Devices compete for channel.
- **Carrier Sensing:** Medium is sensed before transmitting.
- **Collision Handling:** Detection (CSMA/CD) or avoidance (CSMA/CA).
- **Token Passing:** Sequential access control.
- **Bandwidth Utilization:** Varies depending on protocol efficiency.

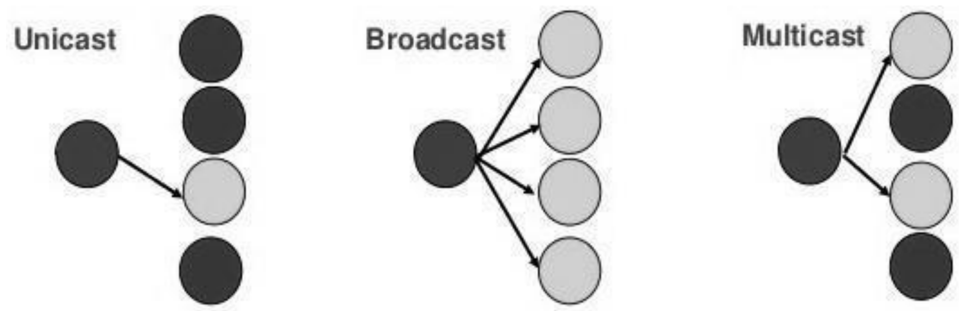
Category	Collision	Delay	Efficiency	Example
Random Access	High	Low	Medium	ALOHA, CSMA
Controlled Access	None	Medium	High	Token Passing
Channelization	None	Low	Very High	FDMA, TDMA, CDMA

LINK-LAYER ADDRESSING

- A link-layer address is sometimes called a link address, sometimes a physical address, and sometimes a MAC address.
- Since a link is controlled at the data-link layer, the addresses need to belong to the data-link layer.
- When a datagram passes from the network layer to the data-link layer, the datagram will be encapsulated in a frame and two data-link addresses are added to the frame header.
- These two addresses are changed every time the frame moves from one link to another.

THREE TYPES OF ADDRESSES

The link-layer protocols define three types of addresses: unicast, multicast, and broadcast.



Unicast Address:

Each host or each interface of a router is assigned a unicast address. Unicasting means one-to-one communication. A frame with a unicast address destination is destined only for one entity in the link.

Multicast Address:

Link-layer protocols define multicast addresses. Multicasting means one-to- many Communication but not all.

Broadcast Address:

Link-layer protocols define a broadcast address. Broadcasting means one- to-all communication. A frame with a destination broadcast address is sent to all entities in the link.

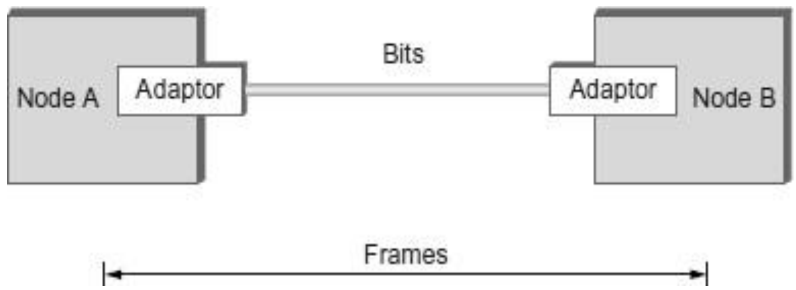
DLC SERVICES

- The data link control (DLC) deals with procedures for communication between two adjacent nodes—node-to-node communication—no matter whether the link is dedicated or broadcast.
- Data link control service include

(1) Framing (2) Flow Control (3) Error Control

1. FRAMING

- The data-link layer packs the bits of a message into frames, so that each frame is distinguishable from another.



- Although the whole message could be packed in one frame, that is not normally done.
- One reason is that a frame can be very large, making flow and error control very inefficient.

- When a message is carried in one very large frame, even a single-bit error would require the retransmission of the whole frame.
- When a message is divided into smaller frames, a single-bit error affects only that small frame.
- Framing in the data-link layer separates a message from one source to a destination by adding a sender address and a destination address.
- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt.

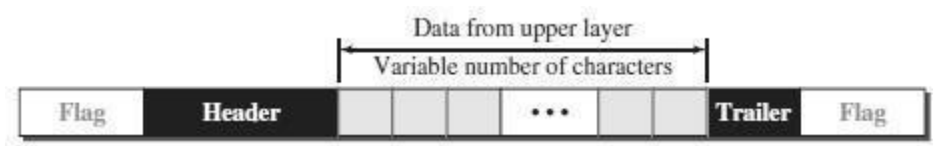
Frame Size

- Frames can be of fixed or variable size.

- Frames of fixed size are called cells. In fixed-size framing, there is no need for defining the boundaries of the frames; the size itself can be used as a delimiter.
- In variable-size framing, we need a way to define the end of one frame and the beginning of the next. Two approaches were used for this purpose: a character-oriented approach and a bit-oriented approach.

Character-Oriented Framing

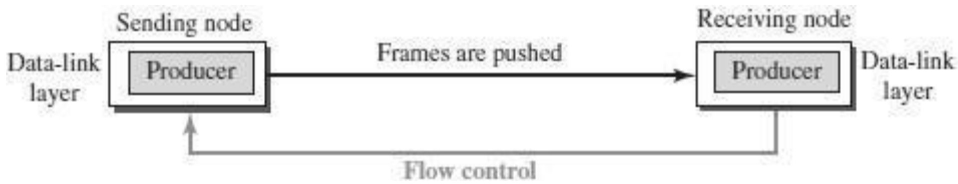
- In character-oriented (or byte-oriented) framing, data to be carried are 8-bit characters.
- To separate one frame from the next, an 8-bit (1-byte) flag is added at the beginning and the end of a frame.
- The flag, composed of protocol-dependent special characters, signals the start or end of a frame.



- Any character used for the flag could also be part of the information.
- If this happens, when it encounters this pattern in the middle of the data,the receiver thinks it has reached the end of the frame.
- To fix this problem, a **byte-stuffing** strategy was added to character- oriented framing.

2. FLOW CONTROL

- **Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.**
- The receiving device has limited speed and limited memory to store the data. o Therefore, the receiving device must be able to inform the sending device to stop the transmission temporarily before the limits are reached. o It requires a buffer, a block of memory for storing the information until they are processed.

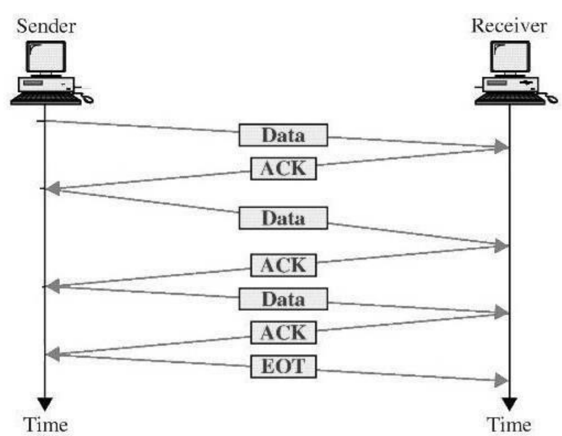


Two methods have been developed to control the flow of data:

- Stop-and-Wait o Sliding Window o

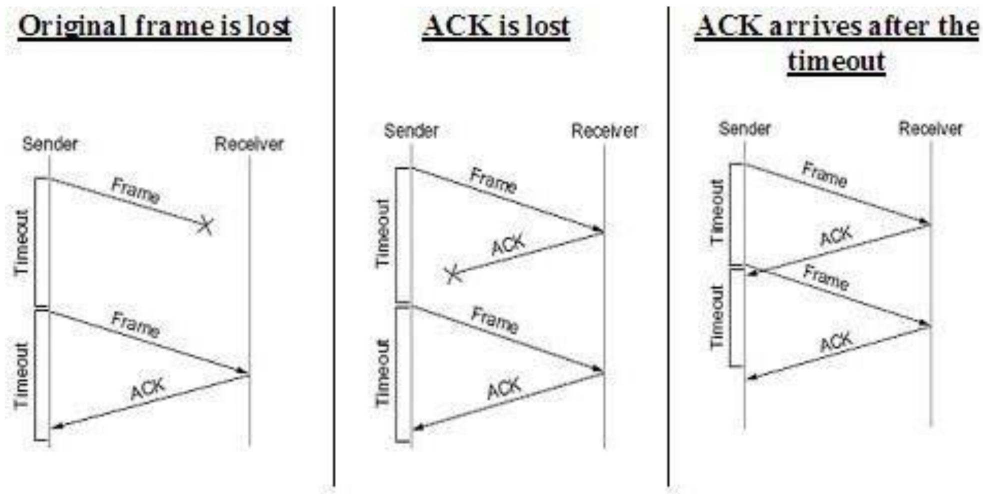
STOP-AND-WAIT

- The simplest scheme is the stop-and-wait algorithm. o In the Stop-and-wait method, the sender waits for an acknowledgement after every frame it sends.
- When acknowledgement is received, then only next frame is sent. o The process of alternately sending and waiting of a frame continues until the sender transmits the EOT (End of transmission) frame.



- If the acknowledgement is not received within the allotted time, then the sender assumes that the frame is lost during the transmission, so it will retransmit the frame.

- The acknowledgement may not arrive because of the following three scenarios :
    1. Original frame is lost
    2. ACK is lost
    3. ACK arrives after the timeout



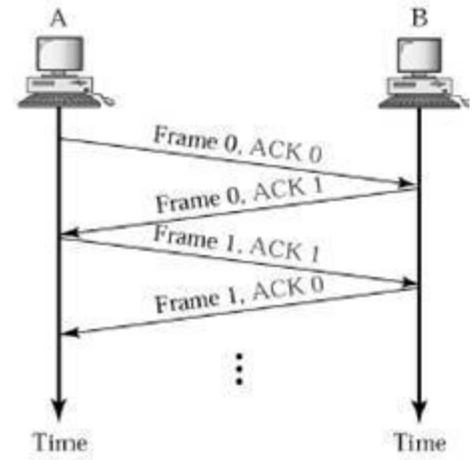
Advantage of Stop-and-wait o The Stop-and-wait method is simple as each frame is checked and acknowledged before the next frame is sent

Disadvantages of Stop-And-Wait o In stop-and-wait, at any point in time, there is only one frame that is sent and waiting to be acknowledged.

- This is not a good use of transmission medium.
- To improve efficiency, multiple frames should be in transition while waiting for ACK.

PIGGYBACKING

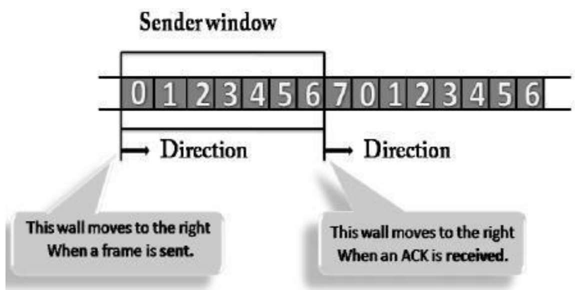
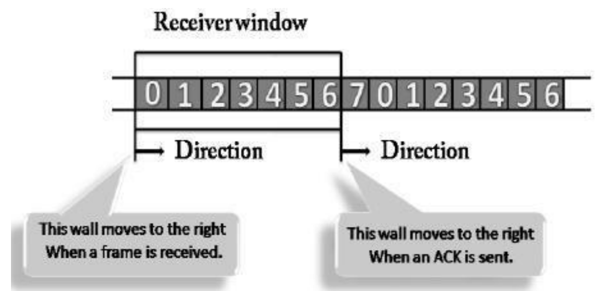
- A method to combine a data frame with ACK.
- Piggybacking saves bandwidth
- Station A and B both have data to send.
- Instead of sending separately, station A sends a data frame that includes an ACK.
- Station B does the same thing.



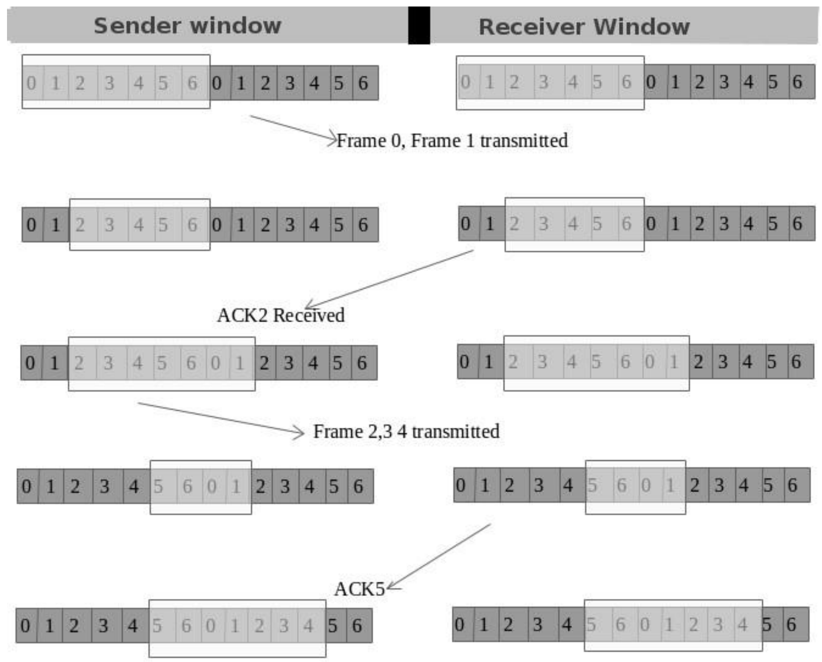
SLIDING WINDOW

- The Sliding Window is a method of flow control in which a sender can transmit the several frames before getting an acknowledgement. ○ In Sliding Window Control, multiple frames can be sent one after the another due to which capacity of the communication channel can be utilized efficiently.
- A single ACK acknowledge multiple frames. ○ Sliding Window refers to imaginary boxes at both the sender and receiver end.
- The window can hold the frames at either end, and it provides the upper limit on the number of frames that can be transmitted before the acknowledgement. ○ Frames can be acknowledged even when the window is not completely filled.
- The window has a specific size in which they are numbered as modulo-n means that they are numbered from 0 to n-1.
- For example, if n = 8, the frames are numbered from 0,1,2,3,4,5,6,7,0,1,2,3,4,5,6,7,0,1........
- The size of the window is represented as n-1. Therefore, maximum n-1 frames can be sent before acknowledgement.
- When the receiver sends the ACK, it includes the number of the next frame that it wants to receive.

- For example, to acknowledge the string of frames ending with frame number 4, the receiver will send the ACK containing the number 5.
- When the sender sees the ACK with the number 5, it got to know that the frames from 0 through 4 have been received.

Sender Window	Receiver Window
	
○ At the beginning of a transmission, the sender window contains n-1 frames. ○ When a frame is sent, the size of the window shrinks. ○ For example, if the size of the window is 'w' and if three frames are sent out, then the number of frames left out in the sender window is w-3. ○ Once the ACK has arrived, then the sender window expands to the number which will be equal to the number of frames acknowledged by ACK.	At the beginning of transmission, the receiver window does not contain n frames, but it contains n-1 spaces for frames. When the new frame arrives, the size of the window shrinks. ○ For example, the size of the window is w and if three frames are received then the number of spaces available in the window is (w-3). Once the acknowledgement is sent, the receiver window expands by the number equal to the number of frames acknowledged.

Example of Sliding Window



4. DATA-LINK LAYER PROTOCOLS

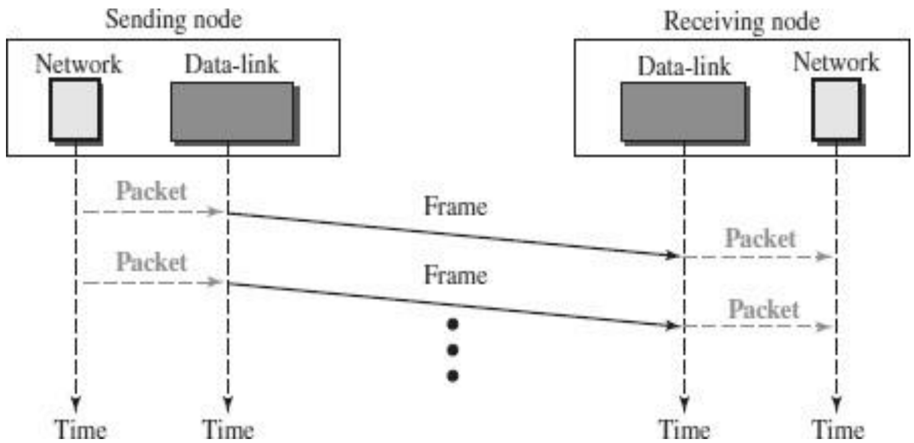
Four protocols have been defined for the data-link layer controls.

They are

1. Simple Protocol
2. Stop-and-Wait Protocol
3. Go-Back-N Protocol
4. Selective-Repeat Protocol

1. SIMPLE PROTOCOL

- The first protocol is a simple protocol with neither flow nor error control.
- We assume that the receiver can immediately handle any frame it receives.
- In other words, the receiver can never be overwhelmed with incoming frames.
- The data-link layers of the sender and receiver provide transmission services for their network layers.



- The data-link layer at the sender gets a packet from its network layer, makes a frame out of it, and sends the frame.
- The data-link layer at the receiver receives a frame from the link, extracts the packet from the frame, and delivers the packet to its network layer.

---

5. <u>HDLC (HIGH-LEVEL DATA LINK CONTROL)</u>

- High-level Data Link Control (HDLC) is a bit-oriented protocol ○ HDLC is used for communication over point-to-point and multipoint links.
- HDLC implements the Stop-and-Wait protocol.

HDLC CONFIGURATIONS AND TRANSFER MODES

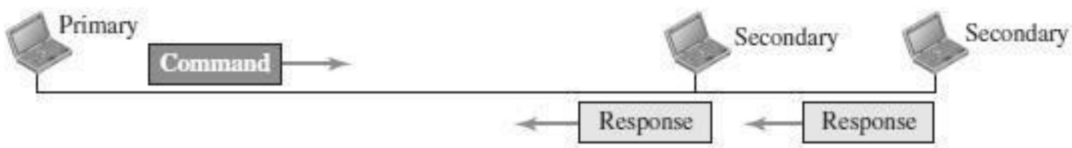
HDLC provides two common transfer modes that can be used in different configurations:

1. Normal response mode (NRM)
2. Asynchronous balanced mode (ABM).

Normal response mode (NRM) ○ In normal response mode (NRM), the station configuration is unbalanced. ○ We have one primary station and multiple secondary stations.

○ A *primary station* can send commands; a *secondary station* can only respond. ○ The NRM is used for both point-to-point and multipoint links.

a. Point-to-point



b. Multipoint

Asynchronous balanced mode (ABM) ○ In ABM, the configuration is balanced. ○ The link is point-to-point, and each station can function as a primary and a secondary (acting as peers). ○ This is the common mode today.

6. POINT-TO-POINT PROTOCOL (PPP)

- Point-to-Point Protocol (PPP) was devised by IETF (Internet Engineering Task Force) in 1990 as a Serial Line Internet Protocol (SLIP).
- PPP is a data link layer communications protocol used to establish a direct connection between two nodes.
- It connects two routers directly without any host or any other networking device in between.
- It is used to connect the Home PC to the server of ISP via a modem.
- It is a byte - oriented protocol that is widely used in broadband communications having heavy loads and high speeds.

- Since it is a data link layer protocol, data is transmitted in frames. It is also known as RFC 1661.

Services Provided by PPP

The main services provided by Point - to - Point Protocol are −

1. Defining the frame format of the data to be transmitted.
2. Defining the procedure of establishing link between two points and exchange of data.
3. Stating the method of encapsulation of network layer data in the frame.
4. Stating authentication rules of the communicating devices.
5. Providing address for network communication.
6. Providing connections over multiple links.
7. Supporting a variety of network layer protocols by providing a range os services.

PPP Frame

PPP is a byte - oriented protocol where each field of the frame is composed of one or more bytes.

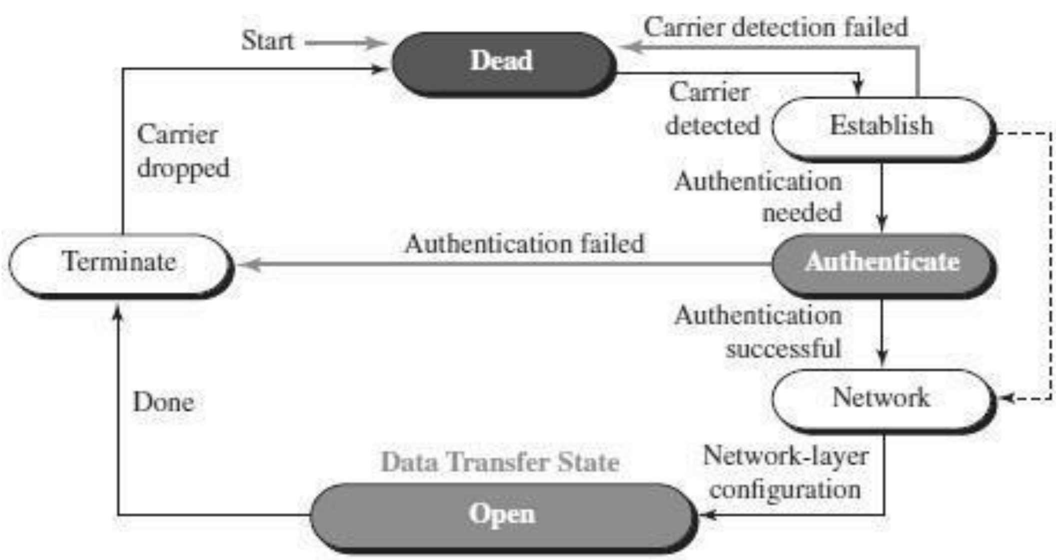
1. **Flag** − 1 byte that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.
2. **Address** − 1 byte which is set to 11111111 in case of broadcast.
3. **Control** − 1 byte set to a constant value of 11000000.
4. **Protocol** − 1 or 2 bytes that define the type of data contained in the payload field.
5. **Payload** − This carries the data from the network layer. The maximum length of the payload field is 1500 bytes.
6. **FCS** − It is a 2 byte(16-bit) or 4 bytes(32-bit) frame check sequence for error detection. The standard code used is CRC.

Byte Stuffing in PPP Frame

Byte stuffing is used is PPP payload field whenever the flag sequence appears in the message, so that the receiver does not consider it as the end of the frame. The escape byte, 01111101, is stuffed before every byte that contains the same byte as the flag byte or the escape byte. The receiver on receiving the message removes the escape byte before passing it onto the network layer.

Transition Phases in PPP

The PPP connection goes through different states as shown in a *transition phase* diagram.



- **Dead**: In dead phase the link is not used. There is no active carrier and the line is quiet.
- **Establish**: Connection goes into this phase when one of the nodes start communication. In this phase, two parties negotiate the options. If negotiation is successful, the system goes into authentication phase or directly to networking phase.
- **Authenticate**: This phase is optional. The two nodes may decide whether they need this phase during the establishment phase. If they decide to proceed with authentication, they send several authentication packets. If the result is successful, the connection goes to the networking phase; otherwise, it goes to the termination phase.

- **Network**: In network phase, negotiation for the network layer protocols takes place. PPP specifies that two nodes establish a network layer agreement before data at the network layer can be exchanged. This is because PPP supports several protocols at network layer. If a node is running multiple protocols simultaneously at the network layer, the receiving node needs to know which protocol will receive the data.
- **Open**: In this phase, data transfer takes place. The connection remains in this phase until one of the endpoints wants to end the connection.
- **Terminate**: In this phase connection is terminated.

Components/Protocols of PPP

Three sets of components/protocols are defined to make PPP powerful:

- Link Control Protocol (LCP)
- Authentication Protocols (AP)
- Network Control Protocols (NCP)

Link Control Protocol (LCP) − It is responsible for establishing, configuring, testing, maintaining and terminating links for transmission. It also provides negotiation mechanisms to set options between the two endpoints. Both endpoints of the link must reach an agreement about the options before the link can be established.

Authentication Protocols (AP) − Authentication means validating the identity of a user who needs to access a set of resources. PPP has created two protocols for authentication -Password Authentication Protocol and Challenge Handshake Authentication Protocol.

PAP

The Password Authentication Protocol (PAP) is a simple authentication procedure with a two-step process:

a. The user who wants to access a system sends an authentication identification (usually the user name) and a password.
b. The system checks the validity of the identification and password and either accepts or denies connection.

CHAP

The Challenge Handshake Authentication Protocol (CHAP) is a three-way handshaking authentication protocol that provides greater security than PAP. In this method, the password is kept secret; it is never sent online.

a. The system sends the user a challenge packet containing a challenge value.
b. The user applies a predefined function that takes the challenge value and the user's own password and creates a result. The user sends the result in the response packet to the system.
c. The system does the same. It applies the same function to the password of the user (known to the system) and the challenge value to create a result. If the result created is the same as the result sent in the response packet, access is granted; otherwise, it is denied.

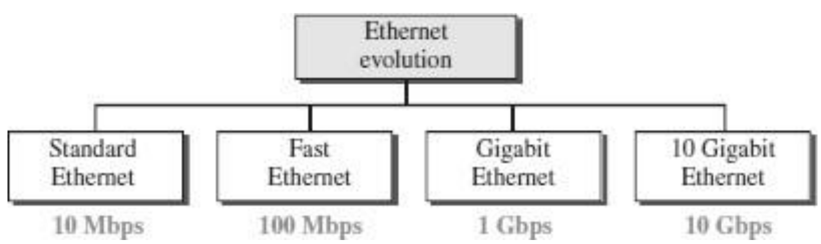
CHAP is more secure than PAP, especially if the system continuously changes the challenge value. Even if the intruder learns the challenge value and the result, the password is still secret.

Network Control Protocols (NCP) – PPP is a multiple-network-layer protocol. It can carry a network-layer data packet from protocols defined by the Internet. PPP has defined a specific Network Control Protocol for each network protocol. These protocols are used for negotiating the parameters and facilities for the network layer. For every higher-layer protocol supported by PPP, one NCP is there.

8. WIRED LAN : ETHERNET (IEEE 802.3)

- Ethernet was developed in the mid-1970's at the Xerox Palo Alto Research Center (PARC),
- IEEE controls the Ethernet standards.
- The Ethernet is the most successful local area networking technology, that uses bus topology.
- The Ethernet is **multiple-access networks** that is set of nodes send and receive frames over a shared link.
- Ethernet uses the **CSMA / CD (C**arrier **S**ense **M**ultiple **A**ccess with **C**ollision **D**etection) mechanism.

EVOLUTION OF ETHERNET



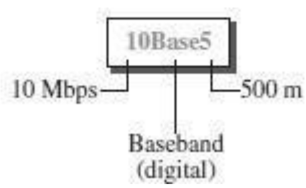
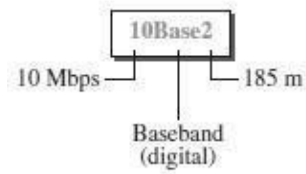
Standard Ethernet (10 Mbps)

The original Ethernet technology with the data rate of 10 Mbps as the Standard Ethernet.

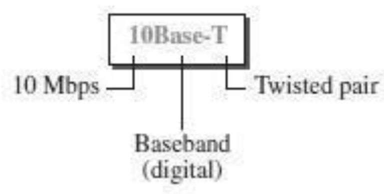
Standard Ethernet types are

1. 10Base5: Thick Ethernet,
2. 10Base2: Thin Ethernet ,

3. 10Base-T: Twisted-Pair Ethernet
4. 10Base-F: Fiber Ethernet.

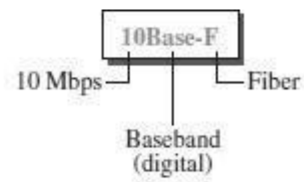
<u>**10Base5: Thick Ethernet**</u>	<u>**10Base2: Thin Ethernet**</u>
 • The first implementation is called ***10Base5, thick Ethernet,*** or ***Thicknet.*** • 10Base5 was the first Ethernet specification to use a bus topology with an external **transceiver**(transmitter/receiver) connected via a tap to a thick coaxial cable.	 The second implementation is called ***10Base2, thin Ethernet,*** or ***Cheapernet.*** 10Base2 also uses a bus topology, but the cable is much thinner and more flexible. In this case, the transceiver is normally part of the network interface card (NIC), which is installed inside the station.

<table>
<tr><th><u>10Base-T: Twisted-Pair Ethernet</u></th><th><u>10Base-F: Fiber Ethernet</u></th></tr>
<tr><td>



- The third implementation is called ***10Base-T*** or ***twisted-pair Ethernet.***
- 10Base-T uses a physical star topology. The stations are connected to a hub via two pairs of twisted cable.

</td><td>



- Although there are several types of optical fiber 10-Mbps Ethernet, the most common is called ***10Base-F.***
- 10Base-F uses a star topology to connect stations to a hub. The stations are connected to the hub using two fiber-optic cables.

</td></tr>
</table>

Fast Ethernet (100 Mbps)

Fast Ethernet or 100BASE-T provides transmission speeds up to 100 megabits per second and is typically used for LAN backbone systems.

The 100BASE-T standard consists of three different component specifications –

1. 100 BASE-TX
2. 100BASE-T4
3. 100BASE-FX

100 BASE-TX	**100BASE-T4**	**100BASE-FX**

100Base-TX uses two pairs of twisted-pair cable either UTP or STP. A 100Base-TX network can provide a data rate of 100 Mbps.	A new standard, called ***100Base-T4,*** was designed to use four pairs of UTP for transmitting 100 Mbps.	**100Base-FX** uses two pairs of fiber-optic cables. Optical fiber can easily handle high bandwidth requirements.

Gigabit Ethernet (1 Gbps)

- The Gigabit Ethernet upgrades the data rate to 1 Gbps(1000 Mbps).
- Gigabit Ethernet can be categorized as either a two-wire or a four-wire implementation.
- The two-wire implementations use fiber-optic cable (**1000Base-SX,** short- wave, or **1000Base-LX,** long-wave), or STP (**1000Base-CX**).
- The four-wire version uses category 5 twisted-pair cable (**1000Base-T**).

10 Gigabit Ethernet(10 Gbps)

- 10 Gigabit Ethernet is an upcoming Ethernet technology that transmits at 10 Gbps.
- 10 Gigabit Ethernet enables a familiar network technology to be used in LAN, MAN and WAN architectures.
- 10 Gigabit Ethernet uses multimode optical fiber up to 300 meters and single mode fiber up to 40 kilometers.
- Four implementations are the most common: **10GBase-SR, 10GBase-LR, 10GBase-EW, and 10GBase-X4.**

ACCESS METHOD/ PROTOCOL OF ETHERNET

The access method of Ethernet is CSMA/CD.

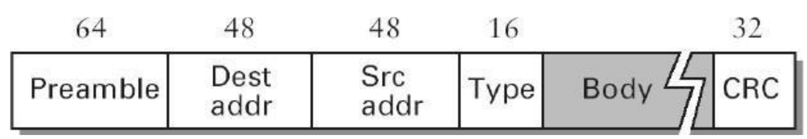
Note : Refer CSMA/CD from MAC

COLLISION DETECTION IN ETHERNET

- As the Ethernet supports collision detection, senders are able to determine a collision.
- At the moment an adaptor detects that its frame is colliding with another, it first makes sure to transmit a **32-bit jamming sequence** along with the **64- bit preamble** (totally 96 bits) and then stops the transmission.
- These **96 bits** are sometimes called **Runt Frame.**

FRAME FORMAT OF ETHERNET

The Ethernet frame is defined by the format given in the Fig.



- The 64-bit ***preamble*** allows the receiver to synchronize with the signal; it is a sequence of alternating 0's and 1's.
- Both the ***source and destination*** hosts are identified with a 48-bit ***address.***
- The packet ***type*** field serves as the demultiplexing key.
- Each frame contains up to 1500 bytes of ***data(Body).***
- ***CRC*** is used for Error detection

Ethernet Addresses

- Every Ethernet host has a unique Ethernet address (48 bits – 6 bytes).
- Ethernet address is represented by sequence of six numbers separated by colons.
- Each number corresponds to 1 byte of the 6 byte address and is given by pair of hexadecimal digits.
- **Eg: 8:0:2b:e4:b1:2** is the representation of
  00001000 00000000 00101011 11100100 10110001 00000010

- Each frame transmitted on an Ethernet is received by every adaptor connected to the Ethernet.
- In addition to ***unicast*** addresses an Ethernet address consisting of ***all 1s*** is treated as ***broadcast*** address.
- Similarly the address that has the ***first bit set to 1*** but it is not the broadcast address is called ***multicast*** address.

9. WIRELESS LAN (IEEE 802.11)

- Wireless communication is one of the fastest-growing technologies.
- The demand for connecting devices without the use of cables is increasing everywhere.
- Wireless LANs can be found on college campuses, in office buildings, and in many public areas.

ADVANTAGES OF WLAN / 802.11

1. **Flexibility:** Within radio coverage, nodes can access each other as radio waves can penetrate even partition walls.
2. **Planning :** No prior planning is required for connectivity as long as devices follow standard convention
3. **Design :** Allows to design and develop mobile devices.
4. **Robustness :** Wireless network can survive disaster. If the devices survive, communication can still be established.

DISADVANTAGES OF WLAN / 802.11

1. **Quality of Service :** Low bandwidth (1 – 10 Mbps), higher error rates due to interference, delay due to error correction and detection.
2. **Cost :** Wireless LAN adapters are costly compared to wired adapters.

3. **Proprietary Solution :** Due to slow standardization process, many solution are proprietary that limit the homogeneity of operation.
4. **Restriction :** Individual countries have their own radio spectral policies. This restricts the development of the technology
5. **Safety and Security :** Wireless Radio waves may interfere with other devices. Eg; In a hospital, radio waves may interfere with high-tech equipment.

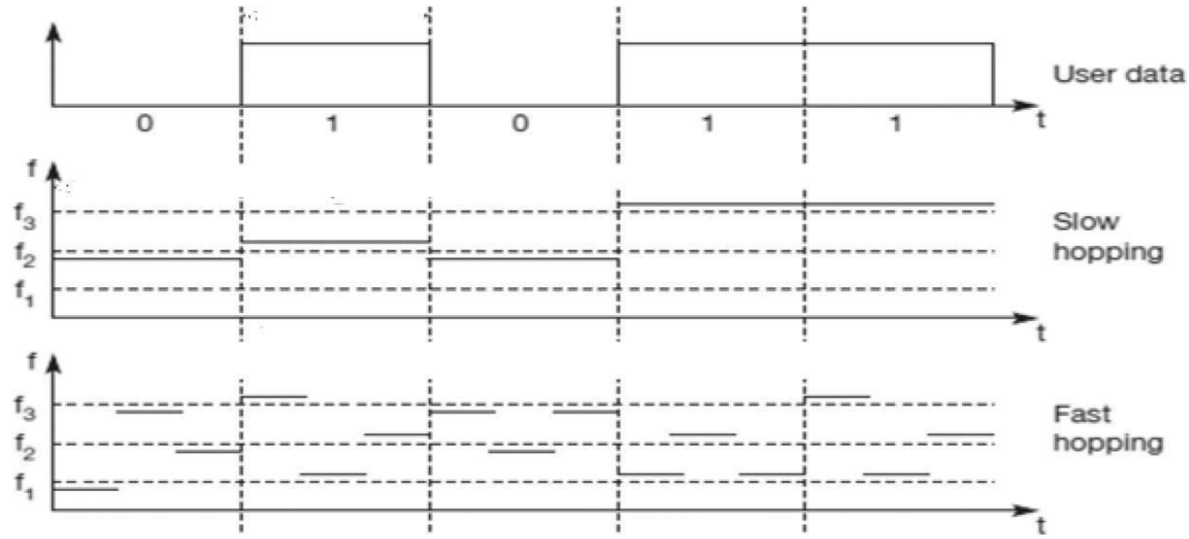
TECHNOLOGY USED IN WLAN / 802.11

- WLAN's uses Spread Spectrum (SS) technology**.**
- The idea behind Spread spectrum technique is to spread the signal over a wider frequency band than normal, so as to minimize the impact of interference from other devices.
- There are two types of Spread Spectrum:

   Frequency Hopping Spread Spectrum (FHSS) Direct **S**equence Spread Spectrum (DSSS)

Frequency Hopping Spread Spectrum (FHSS)

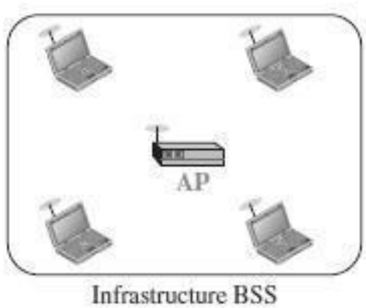
- Frequency hopping is a spread spectrum technique that involves transmitting the signal over a random sequence of frequencies.
- That is, first transmitting at one frequency, then a second, then a third, and so on.
- The random sequence of frequencies is computed by a pseudorandom number generator.
- The receiver uses the same algorithm as the sender and initializes it with the same seed and hence is able to hop frequencies in sync with the transmitter to correctly receive the frame.



TOPOLOGY IN WLAN / 802.11

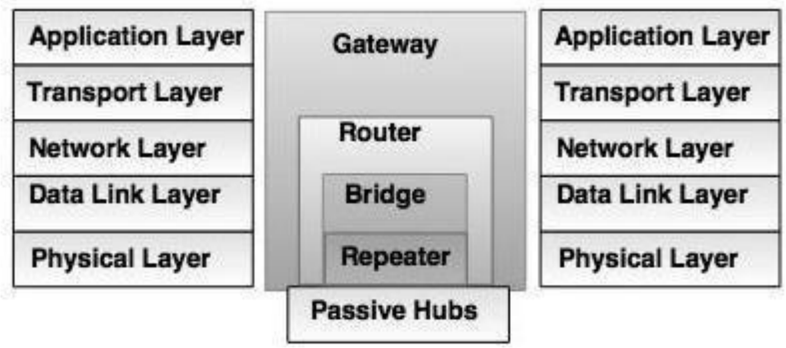
WLANs can be built with either of the following two topologies /architecture:

- Infra-Structure Network Topology
- Ad Hoc Network Topology

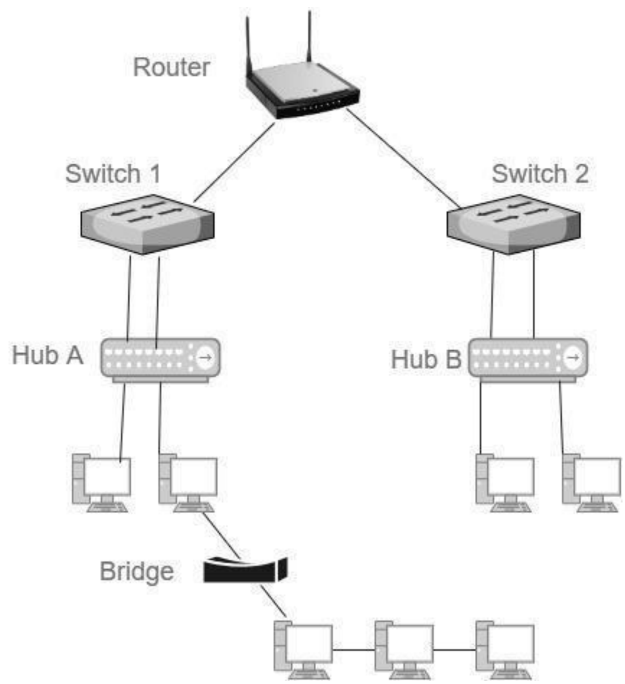
<u>Infra-Structure Topology</u>	**<u>Ad-Hoc Topology</u>** **<u>(Peer-to-Peer Topology)</u>**
 **<u>(AP based Topology)</u>** - An infrastructure network is the network architecture for providing communication between wireless clients and wired network resources. - The transition of data from the wireless to wired medium occurs via a Base Station called AP(Access Point). - An AP and its associated wireless clients define the coverage area.	 - An adhoc network is the architecture that is used to support mutual communication between wireless clients. - Typically, an ad- hoc network is created spontaneously and does not support access to wired networks. - An adhoc network does not require an AP.

11. CONNECTING DEVICES

- Connecting devices are used to connect hosts together to make a network or to connect networks together to make an internet.
- Connecting devices can operate in different layers of the Internet model.



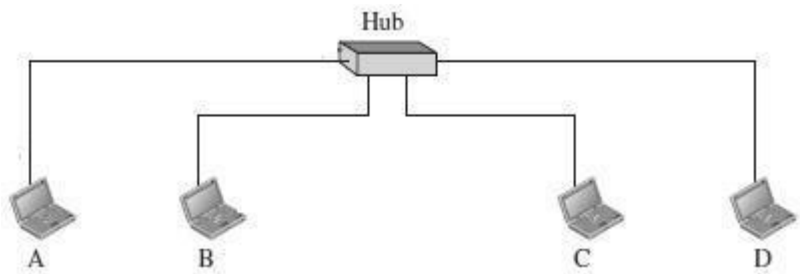
- Connecting devices are divided into five different categories on the basis of layers in which they operate in the network.



1. Devices which operate below the physical layer - **Passive hub**.
2. Devices which operate at the physical layer - **Repeater**.
3. Devices which operate at the physical and data link layers - **Bridge**.
4. Devices which operate at the physical layer, data link layer and network layer – **Router**.
5. Devices which operate at all five layers - **Gateway**.

1. HUBS

- Several networks need a central location to connect media segments together. These central locations are called as hubs.
- The hub organizes the cables and transmits incoming signals to the other media segments.



The three types of hubs are:

i) Passive hub

- It is a connector, which connects wires coming from the different branches.
- By using passive hub, each computer can receive the signal which is sent from all other computers connected in the hub.

ii) Active Hub

- It is a multiport repeater, which can regenerate the signal.
- It is used to create connections between two or more stations in a physical star topology.

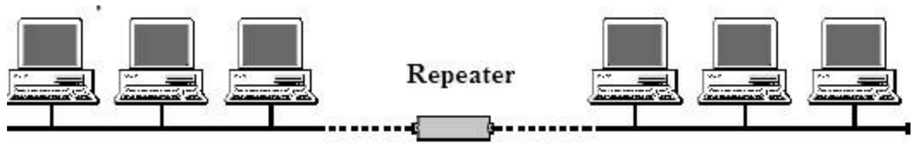
iii) Intelligent Hub

 Intelligent hub contains a program of network management and intelligent path selection.

2. REPEATERS

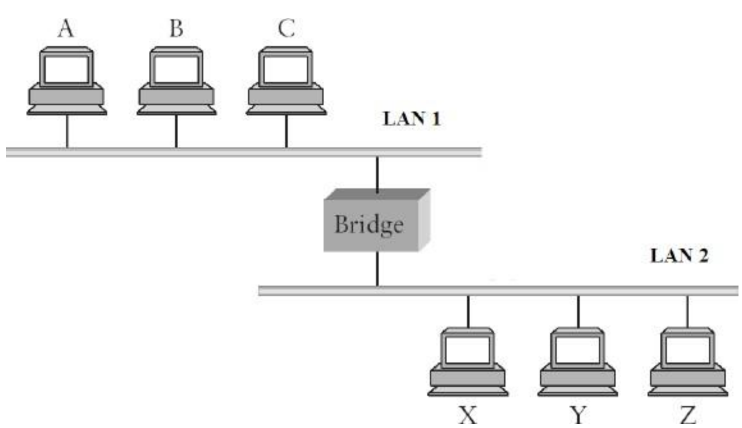
- A repeater receives the signal and it regenerates the signal in original bit pattern before the signal gets too weak or corrupted.
- It is used to extend the physical distance of LAN.
- Repeater works on physical layer.
- A repeater has no filtering capability**.**
- A repeater is implemented in computer networks to expand the coverage area of the network, repropagate a weak or broken signal and or service remote nodes.
- Repeaters amplify the received/input signal to a higher frequency domain so that it is reusable, scalable and available.

- Repeaters are also known as **signal boosters** or **range extender.**
- A repeater cannot connect two LANs, but it connects two segments of the same LAN.



3. BRIDGES

- Bridges operate in physical layer as well as data link layer.
- As a physical layer device, they regenerate the receive signal.
- As a data link layer, the bridge checks the physical (MAC) address (of the source and the destination) contained in the frame.
- The bridge has a filtering feature.
- It can check the destination address of a frame and decides, if the frame should be forwarded or dropped.
- Bridges are used to connect two or LANs working on the same protocol.



Types of Bridges :

➢ Transparent Bridges

These are the bridge in which the stations are completely unaware of the bridge's existence i.e. whether or not a bridge is added or deleted from the network , reconfiguration of the stations is unnecessary.

- Source Routing Bridges

  In these bridges, routing operation is performed by source station and the frame specifies which route to follow.

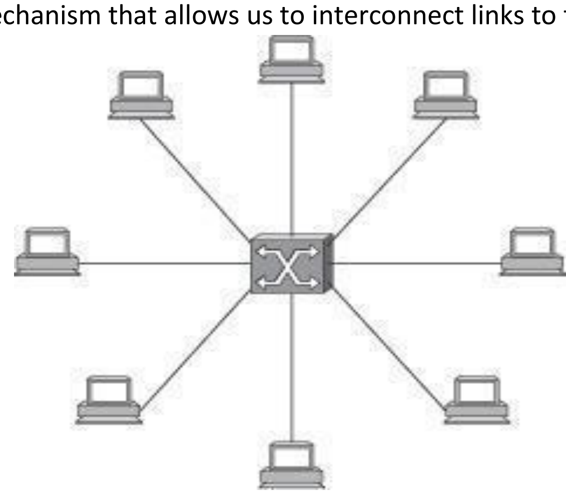
- Translation Bridges

These bridges connect networks with different architectures, such as Ethernet and Token Ring. These bridges appear as:

- Transparent bridges to an Ethernet host
- Source-routing bridges to a Token Ring host

4. SWITCHES

- A switch is a small hardware device which is used to join multiple computers together with one local area network (LAN).
- A switch is a mechanism that allows us to interconnect links to form a large network.



Switch is data link layer device.

A switch is a multi port bridge with a buffer .

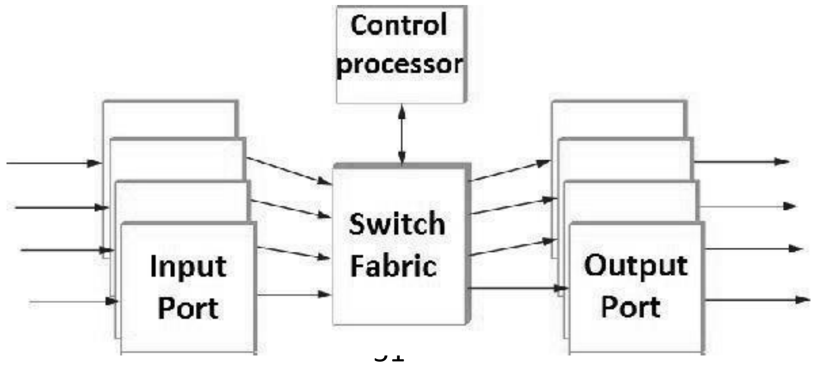
Switches are used to forward the packets based on MAC addresses.

It is operated in full duplex mode.

Packet collision is minimum as it directly communicates between source and destination.

It does not broadcast the message as it works with limited bandwidth. A switch's primary job is to receive incoming packets on one of its links and to transmit them on some other link.

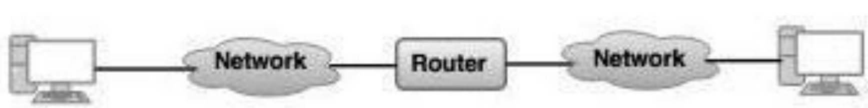
A Switch is used to transfer the data only to the device that has been addressed.



- Input ports receive stream of packets, analyzes the header, determines the output port and passes the packet onto the fabric.
- Ports contain buffers to hold packets before it is forwarded.
- If buffer space is unavailable, then packets are dropped.
- If packets at several input ports queue for a single output port, then only one of them is forwarded.

5. ROUTERS

- A router is a three-layer device.
- It operates in the physical, data-link, and network layers.
- As a physical-layer device, it regenerates the signal it receives.
- As a link-layer device, the router checks the physical addresses (source and destination) contained in the packet.
- As a network-layer device, a router checks the network-layer addresses.
- A router is a device like a switch that routes data packets based on their IP addresses.
- A router can connect networks. A router connects the LANs and WANs on the internet.
- A router is an internetworking device.
- It connects independent networks to form an internetwork.

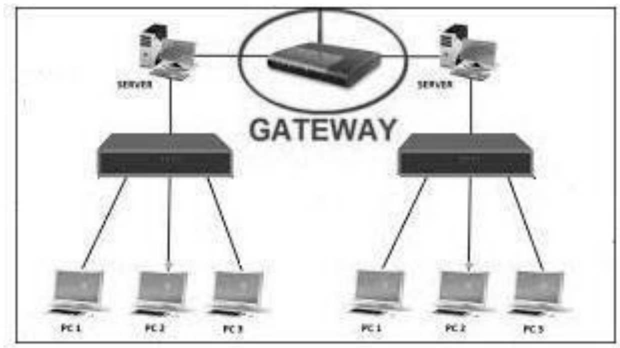


- The key function of the router is to determine the shortest path to the destination.
- Router has a routing table, which is used to make decision on selecting the route.

- The routing table is updated dynamically based on which they make decisions on routing the data packets.

6. GATEWAY

- A gateway is a device, which operates in all five layers of the internet or seven layers of OSI model.
- It is usually a combination of hardware and software. ▪ Gateway connects two independent networks.



- Gateways are generally more complex than switch or router.
- Gateways basically works as the messenger agents that take data from one system, interpret it, and transfer it to another system.
- Gateways are also called protocol converters
- A gateway accepts a packet formatted for one protocol and converts it to a packet formatted to another protocol before forwarding it.
- The gateway must adjust the data rate, size and data format.

7. BROUTER

- Brouter is a hybrid device. It combines the features of both bridge and router.
- Brouter is a combination of Bridge and Router.
- Functions as a bridge for nonroutable protocols and a router for routable protocols.
- As a router, it is capable of routing packets across networks.
- As a bridge, it is capable of filtering local area network traffic.
- Provides the best attributes of both a bridge and a router
- Operates at both the Data Link and Network layers and can replace separate bridges and routers.